4-5-5

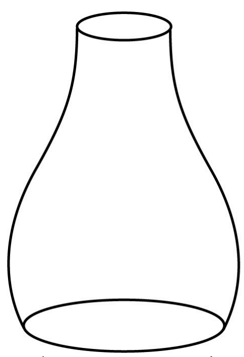
1. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

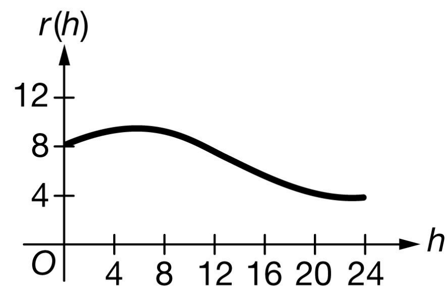
You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.





A 24-centimeter tall vase has cross sections parallel to the base that are circles. Each circular cross section h centimeters above the base has a radius of $r(h) = 8 - 0.1h + 2\sin\left(0.3h^{0.9}\right)$ centimeters for $0 \le h \le 24$. A sketch of the vase and the graph of r are shown above.

(a) Find the area of the region between the graph of r and the h-axis from $h = 0$ to $h = 24$.

(b) Find the volume of the vase.

(c) Evaluate $\int_4^{20} r'(h)\,dh$. Using correct units, interpret the meaning of $\int_4^{20} r'(h)\,dh$ in the context of the problem.

(d) Water is poured into the vase. At the instant when the depth of the water in the vase is 10 centimeters, the depth of the water is increasing at a rate of 0.6 centimeter per second. At that instant, what is the rate of change of the circumference of the surface of the water, in centimeters per second?

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

An unsupported answer of 165.781 (or 165.78) will not earn any points.

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The first point is earned by using $r\ (h)$ as the integrand in a definite integral. A definite integral with incorrect limits or multiplied by any constant other than 1 will not earn the second point.

Do not read units in this problem.

Degree mode: A response that presents answers obtained by using a calculator in degree mode does not earn the first point it would have otherwise earned. The response is generally eligible for all subsequent points (unless no answer is possible in degree mode or the question is made simpler by using degree mode). In degree mode, $\text{area} = 165.509$ (or 165.508).

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Definite integral
- [ ] Answer

Solution:

$$\int_0^{24} r\ (h)\ dh = 165.780506$$

The area of the region is 165.781 (or 165.78).

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for writing a definite integral (with any limits) with integrand $c(r)^2$, for any nonzero constant c.

Once the first point is earned, it cannot be lost. Any subsequent copy errors in the presence of the correct answer will be considered a calculator restart and will earn the second point.

To earn the second point, a response must earn the first point, show limits from 0 to 24, multiply by π, and present the correct evaluation.

Special case: A response that fails to earn the first point because of a copy error will earn the second point for the correct answer.

A final answer written in terms of π must be correct to three places after the decimal point, rounded or truncated, when converted to a decimal value to earn the second point. (For example, 1252.387π is not correct to three digits after the decimal and therefore does not earn the second point.)

Degree mode: $\text{volume} = 3616.746$ (or 3616.745) (See degree mode statement in part (a).)

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0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Integrand
- ☐ Answer

Solution:

$$\int_0^{24} \pi(r\ (h))^2\, dh = \pi \cdot 1252.387790 = 3934.492281$$

The volume of the vase is 3934.492 centimeters.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for the value -5.259; no supporting work is required.

The interpretation must clearly indicate that the radius at $h = 20$ is smaller than the radius at $h = 4$. Therefore, "the difference in the radius when $h = 20$ and $h = 4$ …" is not sufficient.

The response, "The difference (or change) in radius from $h = 4$ to $h = 20$ is -5.259 cm" earns both points. However, "The *total* change in radius from $h = 4$ to $h = 20$ is -5.259 cm" earns only the first point.

Degree mode: answer $= -1.481$ (See degree mode statement in part (a).)

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Value
- ☐ Interpretation with units

Solution:

$$\int_4^{20} r'\ (h)\ dh = r\ (20) - r\ (4) = -5.259388$$

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The radius of the vase at $h = 20$ centimeters is 5.259 centimeters less than the radius of the vase at $h = 4$ centimeters.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned by using $\frac{dr}{dh} \cdot \frac{dh}{dt} \frac{dr}{dt}$ in an expression for $\frac{dC}{dt}$.

A response may work from $C = 2\pi\left(8 - 0.1h + 2\sin\left(0.3h^{0.9}\right)\right)$. In this case, the first point is earned for $\frac{dC}{dt} = \frac{dC}{dh} \cdot \frac{dh}{dt}$.

The second point is earned for using either the value 0.6 as $\frac{dh}{dt}$ or the value -0.411320 as $\frac{dr}{dh}$ in an expression for $\frac{dC}{dt}$. This point is not earned if either 0.6 or $r'(10)$ is used as $\frac{dr}{dt}$.

The second point can be earned without the first point. For example, a computation or communication error in attempting to use the chain rule for $\frac{dC}{dt}$ does not earn the first point but is eligible for the second point.

A response that uses an incorrect equation for the circumference in terms of r only, such as$C = \pi r^2$, is eligible for the first two points but not the third point.

The third point is earned only for an answer of -1.551 (or -1.55).

A final answer written in terms of π must be correct to three places after the decimal point, rounded or truncated, when converted to a decimal value to earn the third point.

Degree mode: $\text{answer} = -0.349$ (or -0.348) (See degree mode statement in part (a).)

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- [ ] $\frac{dr}{dt} = \frac{dr}{dh} \cdot \frac{dh}{dt}$
- [ ] Uses $\left.\frac{dh}{dt}\right|_{h=10} = 0.6$
– OR –
Uses $\left.\frac{dr}{dh}\right|_{h=10} = r'(10)$
- [ ] Answer

Solution:

$$C = 2\pi r \;\Rightarrow\; \frac{dC}{dt} = \frac{d}{dt}(2\pi r) = 2\pi \cdot \left(\frac{dr}{dh} \cdot \frac{dh}{dt}\right)$$

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$$\left.\frac{dC}{dt}\right|_{h=10} = 2\pi \cdot \left(\left.\frac{dr}{dh}\right|_{h=10} \cdot \left.\frac{dh}{dt}\right|_{h=10}\right)$$
$$= 2\pi \cdot (r'(10) \cdot 0.6) = \pi \cdot (-0.493584) = -1.550641$$

The rate of change of the circumference of the surface of the water is -1.551 (or -1.55) centimeters per second.

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2. The wind chill is the temperature, in degrees Fahrenheit (^{o}F), a human feels based on the air temperature, in degrees Fahrenheit, and the wind velocity *v*, in miles per hour (mph). If the air temperature is 32 ^{o}F, then the wind chill is given by $W(v)$=55.6-22.1$v^{0.16}$ and is valid for $5 \leq v \leq 60$.

Over the time interval $0 \leq t \leq 4$ hours, the air temperature is a constant 32^{o}F. At time t=0, the wind velocity is v=20 mph. If the wind velocity increases at a constant rate of 5 mph per hour, what is the rate of change of the wind chill with respect to time at t=3 hours? Indicate units of measure.

Part C

1 point is earned for $\frac{dv}{dt} = 5$

1 point is earned if uses $v(3)$=35 **or** uses $v(5)$=20+5t

1 point is earned for answer

1 point is earned if uses units in (a) and (c)

$$\frac{dW}{dt}\Big|_{t=3} = (\frac{dW}{dv} \cdot \frac{dv}{dt})\Big|_{t=3}$$

=W'(35)*5=-0.892 oF/hr

OR

W=55.6-22.1(20+5t)0.16

$$\frac{dW}{dt}\Big|_{t=3} = -0.892 \text{ oF/hr}$$

Units of oF/mph in (a) and oF/hr in (c)

0	1	2	3	4

The student response earns four of the following points:

1 point is earned for the integrand

1 point is earned if uses $v(3)$=35 **or** uses $v(5)$=20+5t

1 point is earned for answer

1 point is earned if uses units in (a) and (c)

$$\frac{dW}{dt}\Big|_{t=3} = (\frac{dW}{dv} \cdot \frac{dv}{dt})\Big|_{t=3}$$

=W'(35)*5=-0.892 oF/hr

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OR

$W=55.6-22.1(20+5t)0.16$

$\frac{dW}{dt}\Big|_{t=3}=-0.892$ oF/hr

Units of oF/mph in (a) and oF/hr in (c)

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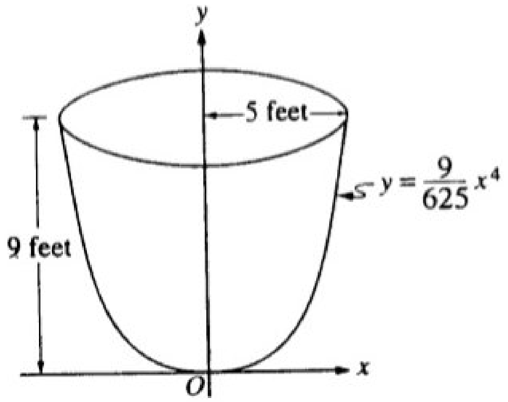
3.



An oil storage tank has the shape as shown above, obtained by revolving the curve $y = \frac{9}{625}x^4$ from $x = 0$ to $x = 5$ about the y-axis, where x and y are measured in feet.

Oil weighing 50 pounds per cubic foot flowed into an initially empty tank at a constant rate of 8 cubic feet per minute. When the depth of oil reached 6 feet, the flow stopped.

Let h be the depth, in feet, of oil in the tank. How fast was the depth of oil in the tank increasing when $h = 4$? Indicate all units of measure.

Part A

1 point is earned for the correct volume as integral with h

$$V = \pi \int_0^h \frac{25}{3}\sqrt{y}dy$$

Finding $\frac{dV}{dt}$:

1 point is earned for the correct $\frac{dV}{dh}$

1 point is earned for the correct $\frac{dV}{dt} = \frac{dV}{dh} \cdot \frac{dh}{dt}$

$$\frac{dV}{dt} = \frac{25\pi}{3}\sqrt{h}\frac{dh}{dt}$$

1 point is earned for the correct answer $\frac{dV}{dt} = 8$

at $h = 4$, $8 = \frac{25\pi}{3}\sqrt{4}\frac{dh}{dt}$

1 point is earned for the correct answer with units

$$\frac{dh}{dt} = \frac{12}{25\pi}\text{ft}/\min$$

Note: If V is linear, max 2/5 (0 0 1 1 0)

If V is constant, max 1/5 (0 0 0 1 0)

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0	1	2	3	4	5

The student response earns five of the following points:

1 point is earned for the correct volume as integral with h

$$V = \pi \int_0^h \frac{25}{3}\sqrt{y}dy$$

Finding $\frac{dV}{dt}$:

1 point is earned for the correct $\frac{dV}{dh}$

1 point is earned for the correct $\frac{dV}{dt} = \frac{dV}{dh} \cdot \frac{dh}{dt}$

$$\frac{dV}{dt} = \frac{25\pi}{3}\sqrt{h}\frac{dh}{dt}$$

1 point is earned for the correct answer $\frac{dV}{dt} = 8$

at $h = 4$, $8 = \frac{25\pi}{3}\sqrt{4}\frac{dh}{dt}$

1 point is earned for the correct answer with units

$$\frac{dh}{dt} = \frac{12}{25\pi}\text{ft}/\text{min}$$

Note: If V is linear, max 2/5 (0 0 1 1 0)

If V is constant, max 1/5 (0 0 0 1 0)

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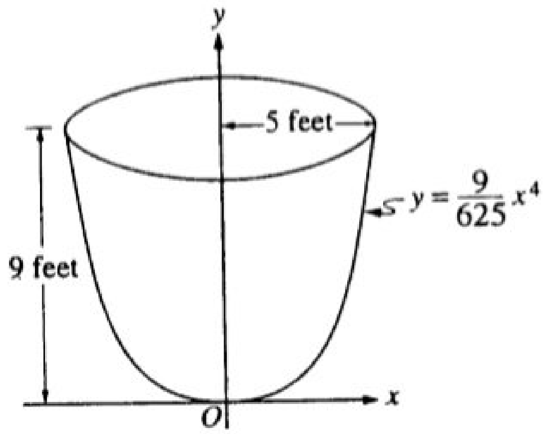
4.



An oil storage tank has the shape as shown above, obtained by revolving the curve $y = \frac{9}{625}x^4$ from x=0 to x=5 about the y-axis, where x and y are measured in feet.

Oil flows into the tank at the constant rate of 8 cubic feet per minute.

Let h be the depth, in feet, of oil in the tank. How fast is the depth of oil in the tank increasing when $h = 4$? Indicate units of measure.

Part C

1 point is earned for the correct volume as definite integral using h

Finding $\frac{dV}{dt}$

$$V = \pi \int_0^h \frac{25}{3}\sqrt{y}dy = \frac{50\pi}{9}h^{3/2}$$

1 point is earned for the correct $\frac{dV}{dh}$

1 point is earned for the correct $\frac{dV}{dt} = \frac{dV}{dh} \cdot \frac{dh}{dt}$ by chain rule

1 point is earned for the correct answer $\frac{dV}{dt} = 8$

$$\frac{dV}{dt} = \frac{25}{3}\pi\sqrt{h}\frac{dh}{dt}$$
$$\frac{dV}{dt} = 8$$

when $h = 4, 8 = \frac{25}{3}\pi(2)\frac{dh}{dt}$

1 point is earned for the correct answer with units

Note: if V is linear, max 2/5 (0 - 0 - 1 - 1 - 0)

if V is constant, max 1/5 (0 - 0 - 0 - 1 - 0)

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$\dfrac{dh}{dt} = \dfrac{12}{25\pi}$ ft/min

(or 0.152 ft/min or 0.153 ft/min)

0	1	2	3	4	5

The student response earns five of the following points:

1 point is earned for the correct volume as definite integral using h

Finding $\dfrac{dV}{dt}$

$$V = \pi \int_0^h \frac{25}{3}\sqrt{y}dy = \frac{50\pi}{9}h^{3/2}$$

1 point is earned for the correct $\dfrac{dV}{dh}$

1 point is earned for the correct by $\dfrac{dV}{dt} = \dfrac{dV}{dh} \cdot \dfrac{dh}{dt}$ chain rule

1 point is earned for the correct answer $\dfrac{dV}{dt} = 8$

$$\frac{dV}{dt} = \frac{25}{3}\pi\sqrt{h}\frac{dh}{dt}$$
$$\frac{dV}{dt} = 8$$

when $h = 4, 8 = \frac{25}{3}\pi(2)\frac{dh}{dt}$

1 point is earned for the correct answer with units

Note: if V is linear, max 2/5 (0 - 0 - 1 - 1 - 0)

if V is constant, max 1/5 (0 - 0 - 0 - 1 - 0)

$\dfrac{dh}{dt} = \dfrac{12}{25\pi}$ ft/min

(or 0.152 ft/min or 0.153 ft/min)

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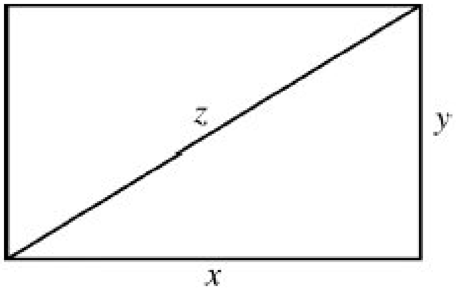
5.



The sides and diagonal of the rectangle above are strictly increasing with time. At the instant when $x=4$ and $y=3$, and $\frac{dx}{dt} = \frac{dz}{dt}$ and $\frac{dy}{dt} = k\frac{dz}{dt}$. What is the value of k at that instant?

(A) $\frac{1}{4}$

(B) $\frac{1}{3}$

(C) 3

(D) 4

(E) It cannot be determined from the information given.

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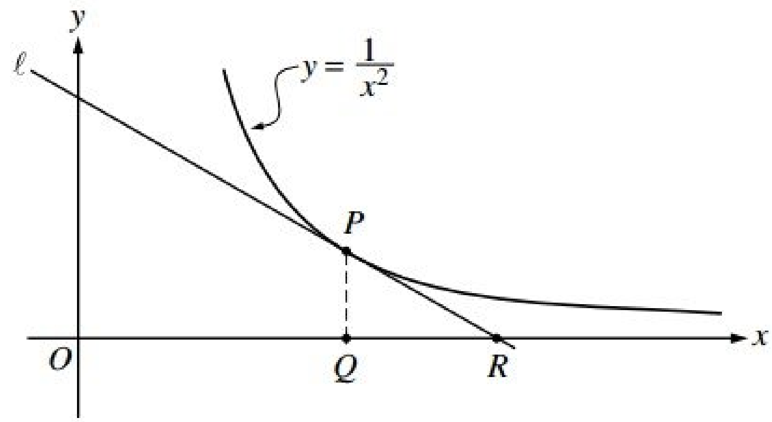
6.



In the figure above, line L is tangent to the graph of $y = \frac{1}{x^2}$ at point P, with coordinates $\left(w, \frac{1}{w^2}\right)$, where $w > 0$. Point Q has coordinates $(w, 0)$. Line L crosses the x-axis at point R, with coordinates $(k, 0)$.

Suppose that w is increasing at the constant rate of 7 units per second. When $w = 5$, what is the rate of change of the area of nPQR with respect to time? Determine whether the area is increasing or decreasing at this instant.

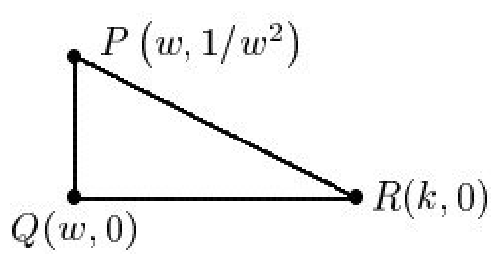
Part D

1 point is earned for the area in term or w and/or k

1 point is earned for: $\frac{dA}{dt}$ implicitly

1 point is earned for: $\left.\frac{dA}{dt}\right|_{w=5}$ using $\frac{dw}{dt} = 7$

1 point is earned for the conclusion



$$A = \tfrac{1}{2}(k - w)\frac{1}{w^2} = \tfrac{1}{2}\left(\tfrac{3}{2}w - w\right)\frac{1}{w^2} = \frac{1}{4w}$$
$$\frac{dA}{dt} = -\frac{1}{4w^2}\frac{dw}{dt}$$
$$\left.\frac{dA}{dt}\right|_{w=5} = -\frac{1}{100} \cdot 7 = -0.07$$

Therefore, area is decreasing.

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0	1	2	3	4

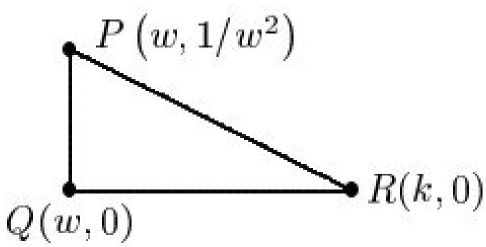
The student response earns all of the following points:

1 point is earned for the area in term or w and/or k

1 point is earned for: $\frac{dA}{dt}$ implicitly

1 point is earned for: $\left.\frac{dA}{dt}\right|_{w=5} u\sin g\frac{dw}{dt} - 7$

1 point is earned for the conclusion



$$A = \frac{1}{2}(k-w)\frac{1}{w^2} = \frac{1}{2}\left(\frac{3}{2}w - w\right)\frac{1}{w^2} = \frac{1}{4w}$$
$$\frac{dA}{dt} = -\frac{1}{4w^2}\frac{dw}{dt}$$
$$\left.\frac{dA}{dt}\right|_{w=5} = -\frac{1}{100}\cdot 7 = -0.07$$

Therefore, area is decreasing.

7. A sphere is expanding in such a way that the area of any circular cross section through the sphere's center is increasing at a constant rate of $2\,cm^2/\text{sec}$. At the instant when the radius of the sphere is 4 centimeters, what is the rate of change of the sphere's volume? (The volume V of a sphere with radius r is given by $V = \frac{4}{3}\pi r^3$.)

(A) $8\,cm^3/\text{sec}$

(B) $16\,cm^3/\text{sec}$ ✓

(C) $8\pi\,cm^3/\text{sec}$

(D) $64\pi\,cm^3/\text{sec}$

(E) $128\pi\,cm^3/\text{sec}$

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8. If the base b of a triangle is increasing at a rate of 3 inches per minute while its height h is decreasing at a rate of 3 inches per minute, which of the following must be true about the area A of the triangle?

(A) A is always increasing.

(B) A is always decreasing.

(C) A is decreasing only when $b<h$.

(D) A is decreasing only when $b>h$. ✓

(E) A remains constant.

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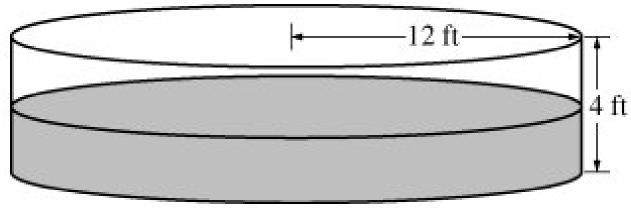
9.

t	0	2	4	6	8	10	12
$P(t)$	0	46	53	57	60	62	63



The figure above shows an above ground swimming pool in the shape of a cylinder with a radius of 12 feet and a height of 4 feet. The pool contains 1000 cubic feet of water at time $t=0$. During the time interval $0 \leq t \leq 12$ hours, water is pumped into the pool at the rate $P(t)$ cubic feet per hour. The table above gives values of $P(t)$ for selected values of t. During the same time interval, water is leaking from the pool at the rate $R(t)$ cubic feet per hour, where $R(t)=25e^{-0.05t}$.

(Note: The volume V of a cylinder with radius r and height h is given by $V=\pi r^2 h$.)

Find the rate at which the volume of water in the pool is increasing at time $t=8$ hours. How fast is the water level in the pool rising at $t=8$ hours? Indicate units of measure in both answers.

Part D

1 point is earned for $V'(8)$

1 point is earned for equation relating $\frac{dv}{dt}$ and $\frac{dh}{dt}$

1 point is earned for $\left.\frac{dh}{dt}\right|_{t=8}$

1 point is earned for units of ft3/ hr and ft/ hr

$V'(t)=P(t)-R(t)$

$$V'(8)=P(8)-R(8)=60-25e^{-0.4} = 43.241 \text{ or } 43.242 \text{ ft}^3/\text{ hr}$$

$$\text{V} = \pi(12)^2 h$$

$$\frac{dV}{dt}=144\pi\frac{dh}{dt}$$

$$\left.\frac{dh}{dt}\right|_{t=8} = \frac{1}{144\pi}\bullet\left.\frac{dV}{dt}\right|_{t=8} = 0.095 \text{ or } 0.096\ ft/hr$$

0	1	2	3	4

The student response earns all of the following points:

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1 point is earned for $V'(8)$

1 point is earned for equation relating $\dfrac{dv}{dt} and \dfrac{dh}{dt}$

1 point is earned for $\left.\dfrac{dh}{dt}\right|_{t=8}$

1 point is earned for units of ft3/ hr and ft/ hr

$V'(t)=P(t)-R(t)$

$V\prime(8) = P(8) - R(8) = 60 - 25\mathrm{e}^{-0.4} = 43.241 \text{ or } 43.242 \text{ ft}^3/\text{ hr}$

$\mathrm{V} = \pi (12)^2 h$

$\dfrac{dV}{dt} = 144\pi\dfrac{dh}{dt}$

$\left.\dfrac{dh}{dt}\right|_{t=8} = \dfrac{1}{144\pi} \bullet \left.\dfrac{dV}{dt}\right|_{t=8} = 0.095\, or\, 0.096\; ft/hr$

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10. At a certain height, a tree trunk has a circular cross section. The radius $R(t)$ of that cross section grows at a rate modeled by the function

$$\frac{dR}{dt} = \frac{1}{16}\left(3 + \sin\left(t^2\right)\right) \text{ centimeters per year for}$$

$0 \le t \le 3$, where time t is measured in years. At time $t = 0$, the radius is 6 centimeters. The area of the cross section at time t is denoted by $A(t)$.

Find the rate at which the cross-sectional area $A(t)$ is increasing at time $t = 3$ years. Indicate units of measure.

Part B

1 point is earned for expression of $A(t)$

1 point is earned for expression of $A'(t)$

1 point is earned for answering with units

$A(t)= \pi\ (R(t))2$

$A'(t)=2\ \pi\ R(t)R'(t)$

A'(3)=8.858 cm2 /year

0	1	2	3

The student response earns all of the following points:

1 point is earned for expression of $A(t)$

1 point is earned for expression of $A'(t)$

1 point is earned for answering with units

$A(t)= \pi\ (R(t))2$

$A'(t)=2\ \pi\ R(t)R'(t)$

A'(3)=8.858 cm2 /year

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11. The fuel consumption of a car, in miles per gallon (mpg), is modeled by $F(s) = 6e^{\left(\frac{s}{20} - \frac{s^2}{2400}\right)}$, where s is the speed of the car, in miles per hour. If the car is traveling at 50 miles per hour and its speed is changing at the rate of 20 miles/hour^2 what is the rate at which its fuel consumption is changing?

(A) 0.215 mpg per hour

(B) 4.299 mpg per hour ✓

(C) 19.793 mpg per hour

(D) 25.793 mpg per hour

(E) 515.855 mpg per hour

12. Functions w, x, and y are differentiable with respect to time and are related by the equation $w = x^2y$. If x is decreasing at a constant rate of 1 unit per minute and y is increasing at a constant rate of 4 units per minute, at what rate is w changing with respect to time when $x = 6$ and $y = 20$?

(A) -384

(B) -240

(C) -96 ✓

(D) 276

(E) 384

13. Sand is deposited into a pile with a circular base. The volume V of the pile is given by $V = \frac{r^3}{3}$, where r is the radius of the base, in feet. The circumference of the base is increasing at a constant rate of 5π feet per hour. When the circumference of the base is 8π feet, what is the rate of change of the volume of the pile, in cubic feet per hour?

(A) $\frac{8}{\pi}$

(B) 16

(C) 40 ✓

(D) 40π

(E) 80π

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14. Oil is leaking from a pipeline on the surface of a lake and forms an oil slick whose volume increases at a constant rate of 2000 cubic centimeters per minute. The oil slick takes the form of a right circular cylinder with both its radius and height changing with time.(Note: The volume V of a right circular cylinder with radius r and height h is given by $V = \pi r^2 h$.)

At the instant when the radius of the oil slick is 100 centimeters and the height is 0.5 centimeter, the radius is increasing at the rate of 2.5 centimeters per minute. At this instant, what is the rate of change of the height of the oil slick with respect to time, in centimeters per minute?

Part A

The response can earn up to 4 points:

1 point: For $\frac{dV}{dt} = 2000$ and $\frac{dr}{dt} = 2.5$

When $r = 100$ cm and $h = 0.5$ cm, $\frac{dV}{dt} = 2000$ cm3/min and $\frac{dr}{dt} = 2.5$ cm/min.

2 point: For the correct expression of $\frac{dV}{dt}$

$$\frac{dV}{dt} = 2\pi r \frac{dr}{dt} h + \pi r^2 \frac{dh}{dt}$$

1 point: For the correct answer

$$2000 = 2\pi(100)(2.5)(0.5) + \pi(100)^2 \frac{dh}{dt} \frac{dh}{dt} = 0.038 \text{ or } 0.039 \text{ cm/min}$$

0	1	2	3	4

The response earns all four of the following points:

1 point: For $\frac{dV}{dt} = 2000$ and $\frac{dr}{dt} = 2.5$

When $r = 100$ cm and $h = 0.5$ cm, $\frac{dV}{dt} = 2000$ cm3/min and $\frac{dr}{dt} = 2.5$ cm/min

2 point: For the correct expression of $\frac{dV}{dt}$

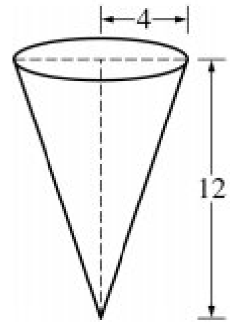
$$\frac{dV}{dt} = 2\pi r \frac{dr}{dt} h + \pi r^2 \frac{dh}{dt}$$

1 point: For the correct answer

$$2000 = 2\pi(100)(2.5)(0.5) + \pi(100)^2 \frac{dh}{dt} \frac{dh}{dt} = 0.038 \text{ or } 0.039 \text{ cm/min}$$

4-5-5

15.



A container has the shape of an open right circular cone, as shown in the figure above. The container has a radius of 4 feet at the top, and its height is 12 feet. If water flows into the container at a constant rate of 6 cubic feet per minute, how fast is the water level rising when the height of the water is 5 feet? (The volume V of a cone with radius r and height h is $V = \frac{1}{3}\pi r^2 h$.)

(A) 0.358 ft/min

(B) 0.688 ft/min

(C) 2.063 ft/min

(D) 8.727 ft/min

(E) 52.360 ft/min

4-5-5

16. The volume of a sphere is increasing at a rate of 6π cubic centimeters per hour. At what rate, in centimeters per hour, is its diameter increasing with respect to time at the instant the radius of the sphere is 3 centimeters?

(Note: The volume of a sphere with radius r is given by $V = \frac{4}{3}\pi r^3$.)

(A) 1/3 ✓

(B) 1

(C) $\sqrt{6}$

(D) 6

Answer A

This option is correct.

Let D = 2r be the diameter of the sphere. This is a related rates question to calculate $\frac{dD}{dt}$ at the moment when r = 3. We are given that $\frac{dV}{dt} = 6\pi$. Using the chain rule:

$$V = \frac{4}{3}\pi r^3 \Rightarrow \frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}.$$

Now substitute the values for $\frac{dV}{dt}$ and r and solve for $\frac{dr}{dt}$.

$$6\pi = 4\pi(3)^2 \frac{dr}{dt} \Rightarrow \frac{dr}{dt} = \frac{1}{6}$$

Since D = 2r, $\frac{dD}{dt} = 2\frac{dr}{dt} = \frac{1}{3}$.

17. The positive variables p and c change with respect to time t. The relationship between p and c is given by the equation $p^2 = (20 - c)^3$.

At the instant when $\frac{dp}{dt} = 41$ and $c = 15$ what is the value of $\frac{dc}{dt}$

(A) $-\frac{82}{75}$

(B) $-\frac{2\sqrt{5}}{3}$

(C) $-\frac{3\sqrt{5}}{2}$

(D) $-\frac{82\sqrt{5}}{15}$ ✓

4-5-5

18. The volume of a certain cone for which the sum of its radius, r, and height is constant is given by $V = \frac{1}{3}\pi r^2 (10 - r)$. The rate of change of the radius of the cone with respect to time is 6. In terms of r, what is the rate of change of the volume of the cone with respect to time?

(A) $-24\pi r$

(B) $6\pi r$

(C) $\frac{20}{3}\pi r - \pi r^2$

(D) $16\pi r - \frac{4}{3}\pi r^2$

(E) $40\pi r - 6\pi r^2$ ✓

19. When x=8, the rate at which $\sqrt[3]{x}$ is increasing is $\frac{1}{k}$ times the rate at which x is increasing. What is the value of k ?

(A) 3

(B) 4

(C) 6

(D) 8

(E) 12 ✓

20. When the area in square units of an expanding circle is increasing twice as fast as its radius in linear units, the radius is

(A) $14/\pi$

(B) 1/4

(C) $1/\pi$ ✓

(D) 1

(E) π

4-5-5

21. A cylindrical can of radius 10 millimeters is used to measure rainfall in Stormville. The can is initially empty, and rain enters the can during a 60-day period. The height of water in the can is modeled by the function S, where $S(t)$ is measured in millimeters and t is measured in days for $0 \leq t \leq 60$. The rate at which the height of the water is rising in the can is given by $S'(t)=2\sin(0.03t)+1.5$.

Assuming no evaporation occurs, at what rate is the volume of water in the can changing at time t=7? Indicate units of measure.

Part C

1 point is earned for relationship between V and S

1 point is earned for the answer

1 point earned for units in (b) or (c).

$$V(t) = 100\pi \mathrm{S}(t)$$
$$V'(7) = 100\pi \mathrm{S}'(7) = 602.218$$

The volume of water in the can is increasing at a rate of $602.218\ \mathrm{mm}^3/day$.

0	1	2	3

The student response all of the following points:

1 point is earned for relationship between V and S

1 point is earned for the answer

1 point earned for units in (b) or (c).

$$V(t) = 100\pi \mathrm{S}(t)$$
$$V'(7) = 100\pi \mathrm{S}'(7) = 602.218$$

The volume of water in the can is increasing at a rate of $602.218\ \mathrm{mm}^3/day$.

22. A person 2 meters tall walks directly away from a streetlight that is 8 meters above the ground. If the person is walking at a constant rate and the person's shadow is lengthening at the rate of 4/9 meter per second, at what rate, in meters per second, is the person walking?

(A) 4/27

(B) 4/9

(C) 3/4

(D) 4/3

(E) 16/9

4-5-5

23. An isosceles right triangle with legs of length s has area $A = \frac{1}{2}s^2$. At the instant when $s = \sqrt{32}$ centimeters, the area of the triangle is increasing at a rate of 12 square centimeters per second.

At what rate is the length of the hypotenuse of the triangle increasing, in centimeters per second, at that instant?

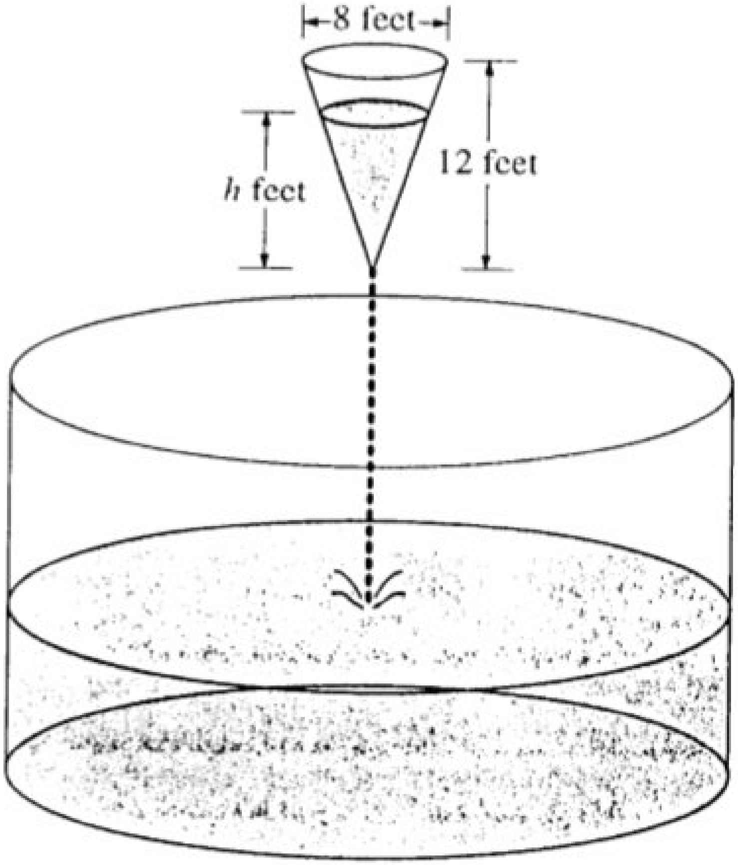
(A) $\frac{3}{4}$

(B) 3 ✓

(C) $\sqrt{32}$

(D) 48

4-5-5



As shown in the figure above, water is draining from a conical tank with height 12 feet and diameter 8 feet into a cylindrical tank that has a base with area 400π square feet. The depth h, in feet, of the water in the conical tank is changing at the rate of $(h-12)$ feet per minute. (The volume V of a cone with radius r and height h is $V = \frac{1}{3}\pi r^2 h$.)

24. At what rate is the volume of water in the conical tank changing when $h = 3$? Indicate units of measure.

Part B

1 point is earned for correctly solving $\frac{dV}{dt}$ using the chain rule

$$\frac{dV}{dt} = \frac{\pi h^2}{9}\frac{dh}{dt}$$

1 point is earned for the correct answer with $\frac{dh}{dt} = h - 12$

$$\frac{\pi h^2}{9}(h - 12) = -9\pi$$

1 point is earned for correctly solving for $\frac{dV}{dt}$ and gives answer with unit

V is decreasing at 9π ft3/min

Note: 0/1 if $\frac{dV}{dt} > 0$

4-5-5

0	1	2	3

The student response earns none of the following points:

1 point is earned for correctly solving $\frac{dV}{dt}$ using the chain rule

$\frac{dV}{dt} = \frac{\pi h^2}{9}\frac{dh}{dt}$

1 point is earned for the correct answer with $\frac{dh}{dt} = h - 12$

$\frac{\pi h^2}{9}(h - 12) = -9\pi$

1 point is earned for correctly solving for $\frac{dV}{dt}$ and gives answer with unit

V is decreasing at 9π ft3/min

Note: 0/1 if $\frac{dV}{dt} > 0$

4-5-5

25. Let y be the depth, in feet, of the water in the cylindrical tank. At what rate is y changing when ? Indicate units of measure.

$h = 3$

Part C

1 point is earned for correctly defining W as a function of y

Let W = volume of water in cylindrical tank

$$W = 400\pi y; \frac{dW}{dt} = 400\pi \frac{dy}{dt}$$
$$400\pi \frac{dy}{dt} = 9\pi$$

1 point is earned for the correct answer for $\frac{dW}{dt} = 400\pi \frac{dy}{dt}$

1 point is earned for correctly identifying $\frac{dW}{dt} = \left|\frac{dV}{dt}\right|$ or -$\frac{dV}{dt}$

1 point is earned for correctly solving for $\frac{dy}{dt}$ and gives answer with units

y is increasing at $\frac{9}{400}$ ft/min

0	1	2	3	4

The student response earns four of the following points:

1 point is earned for correctly defining W as a function of y

Let W = volume of water in cylindrical tank

$$W = 400\pi y; \frac{dW}{dt} = 400\pi \frac{dy}{dt}$$
$$400\pi \frac{dy}{dt} = 9\pi$$

1 point is earned for the correct answer for $\frac{dW}{dt} = 400\pi \frac{dy}{dt}$

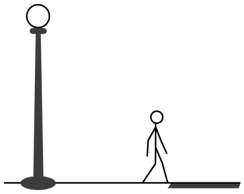
1 point is earned for correctly identifying $\frac{dW}{dt} = \left|\frac{dV}{dt}\right|$ or -$\frac{dV}{dt}$

1 point is earned for correctly solving for $\frac{dy}{dt}$ and gives answer with units

4-5-5

y is increasing at $\frac{9}{400}$ ft/min

26.



A person whose height is 6 feet is walking away from the base of a streetlight along a straight path at a rate of 4 feet per second. If the height of the streetlight is 15 feet, what is the rate at which the person's shadow is lengthening?

(A) 1.5 ft/sec

(B) 2.667 ft/sec ✓

(C) 3.75 ft/sec

(D) 6 ft/sec

(E) 10 ft/sec

27. The volume of a sphere is decreasing at a constant rate of 3 cubic centimeters per second. At the instant when the radius of the sphere is decreasing at a rate of 0.25 centimeter per second, what is the radius of the sphere?

(The volume V of a sphere with radius r is $V = \frac{4}{3}\pi r^3$.)

(A) 0.141 cm

(B) 0.244 cm

(C) 0.250 cm

(D) 0.489 cm

(E) 0.977 cm ✓

28. The radius of a circle is increasing at a constant rate of 0.2 meters per second. What is the rate of increase in the area of the circle at the instant when the circumference of the circle is 20π meters?

(A) 0.04π m^2/sec

(B) 0.4π m^2/sec

(C) 4π m^2/sec ✓

(D) 20π m^2/sec

(E) 100π m^2/sec

4-5-5

29. At time $t \geq 0$, a cube has volume $V(t)$ and edges of length $x(t)$. If the volume of the cube decreases at a rate proportional to its surface area, which of the following differential equations could describe the rate at which the volume of the cube decreases?

(A) $\frac{dV}{dt} = -1.2x^2$ ✓

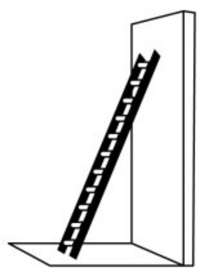
(B) $\frac{dV}{dt} = -1.2x^3$

(C) $\frac{dV}{dt} = -1.2x^2 t$

(D) $\frac{dV}{dt} = -1.2t^2$

(E) $\frac{dV}{dt} = -1.2V^2$

30.



The top of a 15-foot-long ladder rests against a vertical wall with the bottom of the ladder on level ground, as shown above. The ladder is sliding down the wall at a constant rate of 2 feet per second. At what rate, in radians per second, is the acute angle between the bottom of the ladder and the ground changing at the instant the bottom of the ladder is 9 feet from the base of the wall?

(A) $-\frac{2}{9}$ ✓

(B) $-\frac{1}{6}$

(C) $-\frac{2}{25}$

(D) $\frac{2}{25}$

(E) $\frac{1}{9}$

4-5-5

31. A cup has the shape of a right circular cone. The height of the cup is 12 cm, and the radius of the opening is 3 cm. Water is poured into the cup at a constant rate of 2cm^3/sec What is the rate at which the water level is rising when the depth of the water in the cup is 5 cm? (The volume of a cone of height h and radius r is given by $V = \frac{1}{3}\pi r^2 h$)

(A) $\frac{32}{25\pi}$ cm/sec ✓

(B) $\frac{96}{125\pi}$ cm/sec

(C) $\frac{2}{3\pi}$ cm/sec

(D) $\frac{2}{9\pi}$ cm/sec

(E) $\frac{1}{200\pi}$ cm/sec

32. The radius of a circle is decreasing at a constant rate of 0.1 centimeter per second. In terms of the circumference C, what is the rate of change of the area of the circle, in square centimeters per second?

(A) $-(0.2)\,\pi\, C$

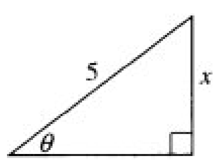
(B) $-(0.1)C$ ✓

(C) $-\frac{(0.1)C}{2\pi}$

(D) $(0.1)^2C$

(E) $(0.1)^2\,\pi C$

33.



In the triangle shown above, if θ increases at a constant rate of 3 radians per minute, at what rate is x increasing in units per minute when x equals 3 units?

(A) 3

(B) $\frac{15}{4}$

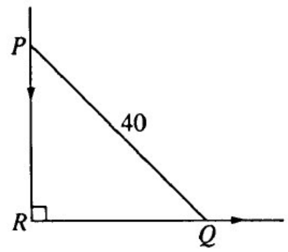
(C) 4

(D) 9

(E) 12 ✓

4-5-5

34.



In the figure above, PQ represents a 40-foot ladder with end P against a vertical wall and end Q on level ground. If the ladder is slipping down the wall, what is the distance RQ at the instant when Q is moving along the ground $\frac{3}{4}$ as fast as P is moving down the wall?

(A) $\frac{6}{5}\sqrt{10}$

(B) $\frac{8}{5}\sqrt{10}$

(C) $\frac{80}{\sqrt{7}}$

(D) 24

(E) 32 ✓

35. The radius of a circle is increasing. At a certain instant, the rate of increase in the area of the circle is numerically equal to twice the rate of increase in its circumference. What is the radius of the circle at that instant?

(A) $\frac{1}{2}$

(B) 1

(C) $\sqrt{2}$

(D) 2 ✓

(E) 4

36. The radius of a circle is increasing at a nonzero rate, and at a certain instant, the rate of increase in the area of the circle is numerically equal to the rate of increase in its circumference. At this instant, the radius of the circle is

(A) $1/\pi$

(B) $1/2$

(C) $2/\pi$

(D) 1 ✓

(E) 2

4-5-5

37. The top of a 25-foot ladder is sliding down a vertical wall at a constant rate of 3 feet per minute. When the top of the ladder is 7 feet from the ground, what is the rate of change of the distance between the bottom of the ladder and the wall?

(A) $-\frac{7}{8}$ feet per minute

(B) $-\frac{7}{24}$ feet per minute

(C) $\frac{7}{24}$ feet per minute

(D) $\frac{7}{8}$ feet per minute ✓

(E) $\frac{21}{25}$ feet per minute

38. The radius of a sphere is decreasing at a rate of 2 centimeters per second. At the instant when the radius of the sphere is 3 centimeters, what is the rate of change, in square centimeters per second, of the surface area of the sphere? (The surface area S of a sphere with radius r is $S = 4\pi r^2$.)

(A) -108π

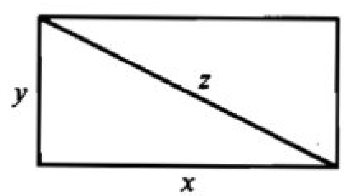
(B) -72π

(C) -48π ✓

(D) -24π

(E) -16π

39.



The sides of the rectangle above increase in such a way that $\frac{dz}{dt} = 1$ and $\frac{dx}{dt} = 3\frac{dy}{dt}$. At the instant when $x = 4$ and $y = 3$, what is the value of $\frac{dx}{dt}$?

(A) $\frac{1}{3}$

(B) 1 ✓

(C) 2

(D) $\sqrt{5}$

(E) 5

4-5-5

40. The volume of a cylinder with radius r inches is given by $V = \pi r^2\left(4r + \sin\left(0.2r^2\right)\right)$. At the instant when the radius is 2.7 inches, the radius is increasing at a rate of 0.75 inch per minute.

At what rate is the volume of the cylinder changing, in cubic inches per minute, at that instant?

(A) 202.575

(B) 220.851

(C) 270.101

(D) 294.467

Answer B

Correct. The rate of change of the volume of the cylinder with respect to time is $\frac{dV}{dt}$. By the chain rule, at the instant when the radius is 2.7 inches and $\frac{dr}{dt} = 0.75$ inch per minute,
$\frac{dV}{dt} = \frac{dV}{dr} \cdot \frac{dr}{dt}\Big|_{r=2.7} = \left(V'(2.7)\,\frac{\text{in}^3}{\text{in}}\right) \cdot \left(0.75\,\frac{\text{in}}{\text{min}}\right) = 220.851\,\frac{\text{in}^3}{\text{min}}$, where the numerical derivative $V'(2.7)$ is evaluated on the graphing calculator.

4-5-5

41. A cylindrical can of radius 10 millimeters is used to measure rainfall in Stormville. The can is initially empty, and rain enters the can during a 60-day period. The height of water in the can is modeled by the function S, where $S(t)$ is measured in millimeters and t is measured in days for $0 \le t \le 60$. The rate at which the height of the water is rising in the can is given by $S'(t) = 2\sin(0.03t) + 1.5$.

(a) According to the model, what is the height of the water in the can at the end of the 60-day period?

(b) According to the model, what is the average rate of change in the height of water in the can over the 60-day period? Show the computations that lead to your answer. Indicate units of measure.

(c) Assuming no evaporation occurs, at what rate is the volume of water in the can changing at time $t = 7$? Indicate units of measure.

(d) During the same 60-day period, rain on Monsoon Mountain accumulates in a can identical to the one in Stormville. The height of the water in the can on Monsoon Mountain is modeled by the function M, where $M(t) = 1/400\ (3t^3 - 30t^2 + 330t)$. The height $M(t)$ is measured in millimeters, and t is measured in days for $0 \le t \le 60$. Let $D(t) = M'(t) - S'(t)$. Apply the Intermediate Value Theorem to the function D on the interval $0 \le t \le 60$ to justify that there exists a time t, $0 < t < 60$, at which the heights of water in the two cans are changing at the same rate.

Part A

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$S(60) = \int_0^{60} S'(t)dt = 171.813 \text{ mm}$$

0	1	2	3

The student response earns all of the following points:

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$S(60) = \int_0^{60} S'(t)dt = 171.813 \text{ mm}$$

Part B

4-5-5

1 point is earned for the answer

1 point earned for units in (b) or (c)

$\frac{S(60)-S(0)}{60} = 2.863 \text{ or } 2.864 \text{ mm/day}$

0	1	2

The student response earns all of the following points:

1 point is earned for the answer

1 point earned for units in (b) or (c)

$\frac{S(60)-S(0)}{60} = 2.863 \text{ or } 2.864 \text{ mm/day}$

Part C

1 point is earned for relationship between V and S

1 point is earned for the answer

1 point earned for units in (b) or (c)

$$V(t) = 100\pi S(t)$$
$$V'(7) = 100\pi S'(7) = 602.218$$

The volume of water in the can is increasing at a rate of $602.218 \text{ mm}^3/\text{day}$.

0	1	2	3

The student response earns all of the following points:

1 point is earned for relationship between V and S

1 point is earned for the answer

1 point earned for units in (b) or (c)

4-5-5

$$V(t) = 100\pi S\,(t)$$
$$V'(7) = 100\pi S'(7) = 602.218$$

The volume of water in the can is increasing at a rate of $602.218\ \text{mm}^3/\text{day}$.

Part D

1 point is earned if considers $D(0)$ and $D(60)$

1 point is earned for justification

$D(0) = -0.675 < 0$ and $D(60) = 69.37730 > 0$

Because D is continuous, the Intermediate Value Theorem implies that there is a time t, $0 < t < 60$, at which $D(t) = 0$. At this time, the heights of water in the two cans are changing at the same rate.

0	1	2 ✓

The student response earns all of the following points:

1 point is earned if considers $D(0)$ and $D(60)$

1 point is earned for justification

$D(0) = -0.675 < 0$ and $D(60) = 69.37730 > 0$

Because D is continuous, the Intermediate Value Theorem implies that there is a time t, $0 < t < 60$, at which $D(t) = 0$. At this time, the heights of water in the two cans are changing at the same rate.

4-5-5

42. A 10-foot ladder is leaning straight up against a wall when a person begins pulling the base of the ladder away from the wall at the rate of 1 foot per second. Which of the following is true about the distance between the top of the ladder and the ground when the base of the ladder is 9 feet from the wall?

(A) The distance is increasing at a rate of $\frac{9}{\sqrt{19}}$ feet per second.

(B) The distance is decreasing at a rate of $\frac{9}{\sqrt{19}}$ feet per second. ✓

(C) The distance is increasing at a rate of $\frac{\sqrt{19}}{9}$ feet per second.

(D) The distance is decreasing at a rate of $\frac{\sqrt{19}}{9}$ feet per second.

Answer B

Correct. Let z denote the distance, in feet, between the top of the ladder and the ground and let y denote the distance, in feet, between the base of the ladder and the wall. Then $10^2 = y^2 + z^2$ and $z = \sqrt{100-81} = \sqrt{19}$ when $y = 9$. Differentiating the first equation with respect to time gives $0 = 2y\frac{dy}{dt} + 2z\frac{dz}{dt}$. Substituting values for y, z, and $\frac{dy}{dt} = 1$, it follows that $0 = (2)(9)(1) + (2)\left(\sqrt{19}\right)\frac{dz}{dt}$. Therefore, $\frac{dz}{dt} = -\frac{9}{\sqrt{19}}$ feet per second. Because $\frac{dz}{dt} < 0$, it follows that the distance is decreasing at this instant.

43. A tube is being stretched while maintaining its cylindrical shape. The height is increasing at the rate of 2 millimeters per second. At the instant that the radius of the tube is 6 millimeters, the volume is increasing at the rate of 96π cubic millimeters per second. Which of the following statements about the surface area of the tube is true at this instant? (The volume V of a cylinder with radius r and height h is $V = \pi r^2 h$. The surface area $S$of a cylinder, not including the top and bottom of the cylinder, is $S = 2\pi rh$.)

(A) The surface area is increasing by 28π square millimeters per second. ✓

(B) The surface area is decreasing by 28π square millimeters per second.

(C) The surface area is increasing by 32π square millimeters per second.

(D) The surface area is decreasing by 32π square millimeters per second.

Answer A

Correct. The goal in this related rates problem is to find $\frac{dS}{dt}$ at the moment when $r = 6$, $\frac{dV}{dt} = 96\pi$, and $\frac{dh}{dt} = 2$.

$96\pi = \frac{dV}{dt} = 2\pi rh\frac{dr}{dt} + \pi r^2\frac{dh}{dt} = 2\pi(6)h\frac{dr}{dt} + \pi(36)(2) \Rightarrow h\frac{dr}{dt} = 2$

$\frac{dS}{dt} = 2\pi\left(r\frac{dh}{dt} + h\frac{dr}{dt}\right) = 2\pi(12+2) = 28\pi$

Because $\frac{dS}{dt} > 0$, the surface area is increasing at this instant.

4-5-5

44. A particle moves on the hyperbola $xy = 18$ for time $t \geq 0$ seconds. At a certain instant, $y = 6$ and $\frac{dy}{dt} = 8$. Which of the following is true about x at this instant?

(A) x is decreasing by 4 units per second.

(B) x is increasing by 4 units per second.

(C) x is decreasing by 1 unit per second.

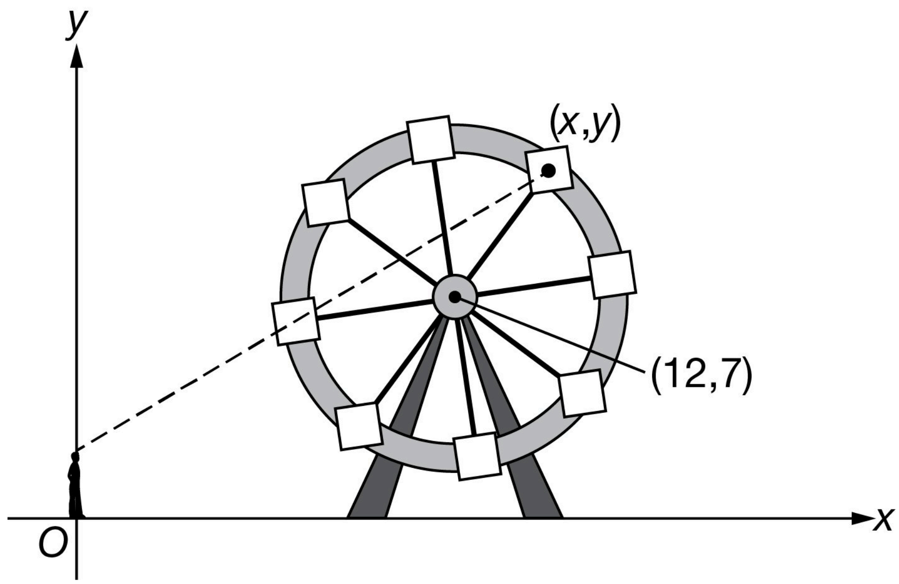
(D) x is increasing by 1 unit per second.

Answer A

Correct. Differentiating both sides of $xy = 18$ with respect to t gives $x\frac{dy}{dt} + y\frac{dx}{dt} = 0$. Substituting the two known values and using $x = \frac{18}{6} = 3$ gives $3(8) + 6\frac{dx}{dt} = 0$. Solving for $\frac{dx}{dt}$ gives $\frac{dx}{dt} = -4$. Because $\frac{dx}{dt} < 0$, x is decreasing at this instant.

4-5-5

45.



The figure above shows a Ferris wheel with radius 5 meters as Jalen, whose eye level is at point $(0, 2)$, watches his friend, Ashanti, ride in one of the cars as the wheel turns. Let Z denote the distance from Jalen to Ashanti's car. The diagram indicates the center of the Ferris wheel at the point $(12, 7)$ and the position of Ashanti's car at the point (x, y). If x and y are functions of time t, in seconds, what is the rate of change of Z when $x = 15$, $y = 11$, and $\frac{dx}{dt} = 1$? (The equation of a circle with radius r and center (h, k) is $(x-h)^2 + (y-k)^2 = r^2$.)

(A) $\frac{dZ}{dt} = -\frac{11}{\sqrt{544}}$, so Ashanti is moving toward Jalen at a rate of approximately 0.47 meter per second.

(B) $\frac{dZ}{dt} = \frac{27}{4\sqrt{346}}$, so Ashanti is moving away from Jalen at a rate of approximately 0.36 meter per second.

(C) $\frac{dZ}{dt} = \frac{11}{\sqrt{544}}$, so Ashanti is moving away from Jalen at a rate of approximately 0.47 meter per second. ✓

(D) $\frac{dZ}{dt} = 13.5$, so Ashanti is moving away from Jalen at a rate of 13.5 meters per second.

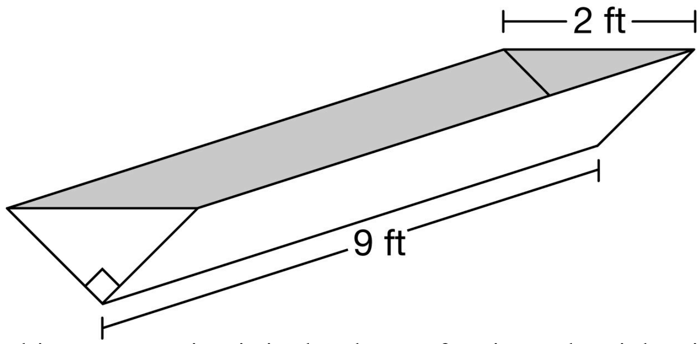
Answer C

Correct. The functions x and y satisfy $(x-12)^2 + (y-7)^2 = 25$, and differentiating gives $2(x-12)\frac{dx}{dt} + 2(y-7)\frac{dy}{dt} = 0$. Substituting the three known values and solving for $\frac{dy}{dt}$ yields $\frac{dy}{dt} = -\frac{3}{4}$. Since $Z = \sqrt{x^2 + (y-2)^2}$, $\frac{dZ}{dt} = \frac{2x\frac{dx}{dt} + 2(y-2)\frac{dy}{dt}}{2\sqrt{x^2+(y-2)^2}}$. Substituting for x, y, $\frac{dx}{dt}$, and $\frac{dy}{dt}$ gives

$\frac{dZ}{dt} = \frac{11\sqrt{34}}{136} = \frac{11}{\sqrt{544}} \approx 0.47$. So Ashanti is moving away from Jalen at approximately 0.47 meter per second.

4-5-5

46.



A trough is 9 feet long, and its cross section is in the shape of an isosceles right triangle with hypotenuse 2 feet, as shown above. Water begins flowing into the empty trough at the rate of 2 cubic feet per minute. At what rate is the height h feet of the water in the trough changing 2 minutes after the water begins to flow?

(A) Decreasing at $\frac{2}{3}$ foot per minute

(B) Increasing at $\frac{2}{3}$ foot per minute

(C) Decreasing at $\frac{1}{6}$ foot per minute

(D) Increasing at $\frac{1}{6}$ foot per minute

Answer D

Correct. The volume $V(t)$ of the water in the trough at time t is $V(t) = 9(h(t))^2$, where $h(t)$ is the height of the water at time t. Water is flowing into the empty trough at the rate of 2 cubic feet per minute, so $V(2) = 4$. It follows that $4 = 9(h(2))^2$, so $h(2) = \frac{2}{3}$. Also, $2 = \frac{dV}{dt} = 18h\frac{dh}{dt} \Rightarrow \frac{dh}{dt} = \frac{1}{9h}$, so when $t = 2$, $\frac{dh}{dt} = \left(\frac{1}{9}\right)\left(\frac{3}{2}\right) = \frac{1}{6}$. Therefore, 2 minutes after water begins to flow in, the height of the water in the trough is increasing at $\frac{1}{6}$ foot per minute.

4-5-5

47. A particle moves on the circle $x^2 + y^2 = 100$ in the xy-plane for time $t \geq 0$. At the time when the particle is at the point $(8, 6)$, the value of $\frac{dx}{dt}$ is 5. What is the value of $\frac{dy}{dt}$ at this time?

(A) $\frac{dy}{dt} = -\frac{20}{3}$

(B) $\frac{dy}{dt} = -\frac{4}{3}$

(C) $\frac{dy}{dt} = \frac{5}{3}$

(D) $\frac{dy}{dt} = 7$

Answer A

Correct. Differentiating both sides of $x^2 + y^2 = 100$ with respect to t gives $2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0$. Substituting in the three known values gives $2(8)(5) + 2(6)\frac{dy}{dt} = 0$. Solving for $\frac{dy}{dt}$ gives the result $\frac{dy}{dt} = -\frac{20}{3}$.

48. A piece of rubber tubing maintains a cylindrical shape as it is stretched. At the instant that the inner radius of the tube is 2 millimeters and the height is 20 millimeters, the inner radius is decreasing at the rate of 0.1 millimeter per second and the height is increasing at the rate of 3 millimeters per second. Which of the following statements about the volume of the tube is true at this instant? (The volume V of a cylinder with radius r and height h is $V = \pi r^2 h$.)

(A) The volume is increasing by 4π cubic millimeters per second.

(B) The volume is decreasing by 4π cubic millimeters per second.

(C) The volume is increasing by 20π cubic millimeters per second.

(D) The volume is decreasing by 20π cubic millimeters per second.

Answer A

Correct. The goal in this related rates problem is to find $\frac{dV}{dt}$ at the moment when $r = 2$, $h = 20$, $\frac{dr}{dt} = -0.1$ (because r is decreasing), and $\frac{dh}{dt} = 3$.

$\frac{dV}{dt} = 2\pi rh\frac{dr}{dt} + \pi r^2\frac{dh}{dt} = 2\pi(2)(20)(-0.1) + \pi(4)(3) = -8\pi + 12\pi = 4\pi$

Because $\frac{dV}{dt} > 0$, the volume is increasing at this instant.

4-5-5

49. A particle moves on the hyperbola $xy = 15$ for time $t \geq 0$ seconds. At a certain instant, $x = 3$ and $\frac{dx}{dt} = 6$. Which of the following is true about y at this instant?

(A) y is decreasing by 10 units per second. ✓

(B) y is increasing by 10 units per second.

(C) y is decreasing by 5 units per second.

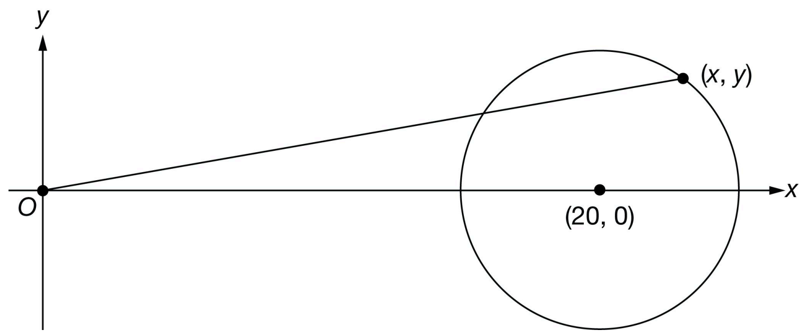
(D) y is increasing by 5 units per second.

Answer A

Correct. Differentiating both sides of $xy = 15$ with respect to t gives $x\frac{dy}{dt} + y\frac{dx}{dt} = 0$. Substituting the two known values and using $y = \frac{15}{3} = 5$ gives $3\frac{dy}{dt} + 5(6) = 0$. Solving for $\frac{dy}{dt}$ gives $\frac{dy}{dt} = -10$. Because $\frac{dy}{dt} < 0$, y is decreasing at this instant.

4-5-5

50.



A model car travels around a circular track with radius 5 feet. Let Z denote the distance between the model car and a fixed point that is 20 feet to the left of the center of the circular track. The diagram above indicates the fixed point at the origin, the center of the circular track at the point $(20, 0)$, and the position of the car at the point (x, y). Z is the length of the line segment from the origin to the point (x, y). If x and y are functions of time t, in seconds, what is the rate of change of Z when $x = 23$, $y = 4$, and $\frac{dx}{dt} = 2$? (The equation of a circle with radius r and center (h, k) is $(x-h)^2 + (y-k)^2 = r^2$.)

(A) $\frac{dZ}{dt} = 0$, so the distance between the model car and the fixed point is constant.

(B) $\frac{dZ}{dt} = \frac{40}{\sqrt{545}}$, so the model car is moving away from the fixed point at a rate of approximately 1.7 feet per second.

(C) $\frac{dZ}{dt} = \frac{40}{\sqrt{545}}$, so the model car is moving toward the fixed point at a rate of approximately 1.7 feet per second.

(D) $\frac{dZ}{dt} = 80$, so the model car is moving away from the fixed point at a rate of 80 feet per second.

Answer B

Correct. The functions x and y satisfy $(x-20)^2 + y^2 = 25$ and differentiating gives $2(x-20)\frac{dx}{dt} + 2y\frac{dy}{dt} = 0$. Substituting the three known values and solving for $\frac{dy}{dt}$ yields $\frac{dy}{dt} = -\frac{3}{2}$. Since $Z = \sqrt{x^2+y^2}$, $\frac{dZ}{dt} = \frac{2x\frac{dx}{dt}+2y\frac{dy}{dt}}{2\sqrt{x^2+y^2}}$. Substituting for x, y, $\frac{dx}{dt}$, and $\frac{dy}{dt}$ gives $\frac{dZ}{dt} = \frac{40}{\sqrt{545}}$.

4-5-5

51. Two straight roads intersect at right angles. A car is traveling south toward the intersection while a radar-monitored-speed sign is positioned $\frac{1}{4}$ mile east of the intersection. The speed sign provides both the distance from the sign to an approaching car and the rate at which the distance between the sign and the car is changing. At a certain time, the speed sign shows that the approaching car is exactly $\frac{1}{2}$ mile away from the sign and that the rate at which the distance between the car and the sign is decreasing is 60 miles per hour. Which of the following is true about the distance between the car and the intersection at this instant?

(A) The distance is increasing at a rate of $\frac{120}{\sqrt{3}}$ miles per hour.

(B) The distance is decreasing at a rate of $\frac{120}{\sqrt{3}}$ miles per hour. ✓

(C) The distance is increasing at a rate of $\frac{30}{\sqrt{3}}$ miles per hour.

(D) The distance is decreasing at a rate of $\frac{30}{\sqrt{3}}$ miles per hour.

Answer B

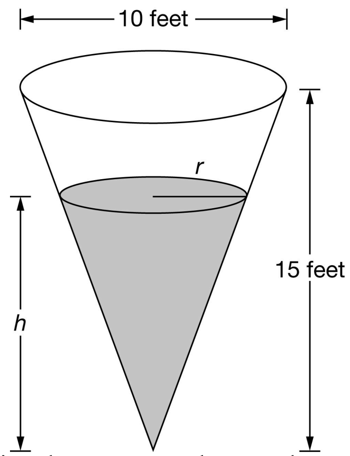
Correct. Let z denote the distance between the car and the speed sign and y denote the distance between the approaching car and the intersection. Then $z^2 = y^2 + \left(\frac{1}{4}\right)^2$ and $y = \sqrt{\frac{1}{4} - \frac{1}{16}} = \frac{\sqrt{3}}{4}$ when $z = \frac{1}{2}$. Differentiating the first equation with respect to time gives $2z\frac{dz}{dt} = 2y\frac{dy}{dt} + 0$. Substituting values for y, z, and $\frac{dz}{dt} = -60$, it follows that $2\left(\frac{1}{2}\right)(-60) = 2\left(\frac{\sqrt{3}}{4}\right)\frac{dy}{dt} + 0$. Therefore,

$\frac{dy}{dt} = \frac{\frac{1}{2}(-60)}{\frac{\sqrt{3}}{4}} = \frac{-120}{\sqrt{3}}$ miles per hour.

Because $\frac{dy}{dt} < 0$, it follows that the distance is decreasing at this instant.

4-5-5

52.



A water tank is in the shape of a right circular cone as shown above. The diameter of the cone is 10 feet, and the height is 15 feet. The shape of the water in the tank is conical with radius r feet and height h feet. At noon, water is leaking from the bottom of the tank at a rate of 12 cubic feet per hour, and the volume of water in the tank is 27π cubic feet. At noon, what is the rate at which the height of the water in the tank is changing? (The volume V of a right circular cone with radius r and height h is $V = \frac{1}{3}\pi r^2 h$.)

(A) Decreasing at $\frac{4}{9\pi}$ feet per hour

(B) Increasing at $\frac{4}{9\pi}$ feet per hour

(C) Increasing at $\frac{4}{3\pi}$ feet per hour

(D) Decreasing at $\frac{4}{3\pi}$ feet per hour ✓

Answer D

Correct. The radius of the cone is 5 feet. Using similarity, it follows that $r = \frac{h}{3}$. Let V denote the volume of the cone of water in the tank. Then $V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi\left(\frac{h}{3}\right)^2 h = \frac{\pi h^3}{27}$. From this equation, it follows that when $V = 27\pi$, $h = 9$. Differentiating with respect to t, $\frac{dV}{dt} = \frac{\pi h^2}{9}\frac{dh}{dt}$. Substituting values into the differential equation we have $-12 = \frac{9^2\pi}{9}\frac{dh}{dt}$ so $\frac{dh}{dt} = -\frac{4}{3\pi}$. Therefore, the rate at which the height of the cone of water in the tank is changing is decreasing at $\frac{4}{3\pi}$ feet per hour.

4-5-5

53. A particle moves on the circle $x^2 + y^2 = 25$ in the xy-plane for time $t \geq 0$. At the time when the particle is at the point $(3, 4)$, $\frac{dx}{dt} = 6$. What is the value of $\frac{dy}{dt}$ at this time?

(A) $\frac{dy}{dt} = \frac{19}{8}$

(B) $\frac{dy}{dt} = -\frac{3}{4}$

(C) $\frac{dy}{dt} = -\frac{11}{8}$

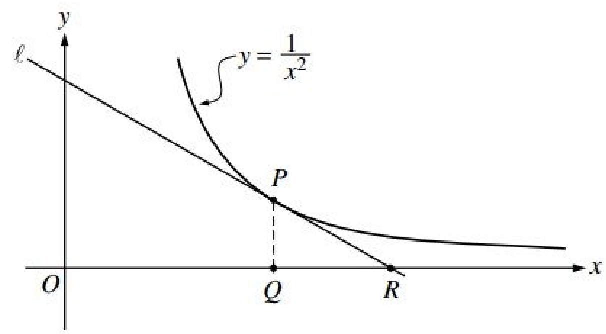
(D) $\frac{dy}{dt} = -\frac{9}{2}$

Answer D

Correct. Differentiating both sides of $x^2 + y^2 = 25$ with respect to t gives $2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0$. Substituting in the three known values gives $2\left(3\right)\left(6\right) + 2\left(4\right)\frac{dy}{dt} = 0$. Solving for $\frac{dy}{dt}$ gives the result $\frac{dy}{dt} = -\frac{9}{2}$.

4-5-5

54.



In the figure above, line ℓ is tangent to the graph of $y = \frac{1}{x^2}$ at point P, with coordinates $\left(w, \frac{1}{w^2}\right)$, where $w > 0$. Point Q has coordinates $(w, 0)$. Line ℓ crosses the x-axis at point R, with coordinates $(k, 0)$.

(a) Find the value of k when $w = 3$.

(b) For all $w > 0$, find k in terms of w.

(c) Suppose that w is increasing at the constant rate of 7 units per second. When $w = 5$, what is the rate of change of k with respect to time?

(d) Suppose that w is increasing at the constant rate of 7 units per second. When $w = 5$, what is the rate of change of the area of n PQR with respect to time? Determine whether the area is increasing or decreasing at this instant.

Part A

1 point is earned for: $\left.\frac{dy}{dx}\right|_{x=3}$

1 point is earned for the answer

$$\frac{dy}{dx} = -\frac{2}{x^3}; \left.\frac{dy}{dx}\right|_{x=3} = -\frac{2}{27}$$

Line ℓ through $\left(3, \frac{1}{9}\right)$and $(k, 0)$ has slope $-\frac{2}{27}$.

Therefore, $\frac{0-\frac{1}{9}}{k-3} = -\frac{2}{27}$ or, $0 - \frac{1}{9} = -\frac{2}{27}(k-3)$

$k = \frac{9}{2}$

4-5-5

0	1	2

The student response earns all of the following points:

1 point is earned for: $\left.\frac{dy}{dx}\right|_{x=3}$

1 point is earned for the answer

$\frac{dy}{dx} = -\frac{2}{x^3}; \left.\frac{dy}{dx}\right|_{x=3} = -\frac{2}{27}$

Line ℓ through $\left(3, \frac{1}{9}\right)$ and $(k, 0)$ has slope $-\frac{2}{27}$.

Therefore, $\frac{0-\frac{1}{9}}{k-3} = -\frac{2}{27}$ or, $0 - \frac{1}{9} = -\frac{2}{27}(k-3)$

$k = \frac{9}{2}$

Part B

1 point is earned for equation relating w and k, using slopes

1 point is earned for the answer

Line ℓ through $\left(w, \frac{1}{w^2}\right)$ and $(k, 0)$ has slope $-\frac{2}{w^3}$.

Therefore, $\frac{0-\frac{1}{w^2}}{k-w} = -\frac{2}{w^3}$ or $0 - \frac{1}{w^2} = -\frac{2}{w^3}(k-w)$

$k = \frac{3}{2}w$

0	1	2

4-5-5

The student response earns all of the following points:

1 point is earned for equation relating w and k, using slopes

1 point is earned for the answer

Line ℓ through $\left(w, \frac{1}{w^2}\right)$ and $(k, 0)$ has slope $-\dfrac{2}{w^3}$.

Therefore, $\frac{0-\frac{1}{w^2}}{k-w} = -\frac{2}{w^3}$ or $0 - \frac{1}{w^2} = -\frac{2}{w^3}(k-w)$
$k = \frac{3}{2}w$

Part C

1 point is earned for the answer using: $\dfrac{dw}{dt} = 7$

$$\frac{dk}{dt} = \frac{3}{2}\frac{dw}{dt} = \frac{3}{2}\cdot 7 = \frac{21}{2};\ \left.\frac{dk}{dt}\right|_{w=5} = \frac{21}{2}$$

0	1

The student response earns all of the following points:

1 point is earned for the answer using: $\dfrac{dw}{dt} = 7$

$$\frac{dk}{dt} = \frac{3}{2}\frac{dw}{dt} = \frac{3}{2}\cdot 7 = \frac{21}{2};\ \left.\frac{dk}{dt}\right|_{w=5} = \frac{21}{2}$$

Part D

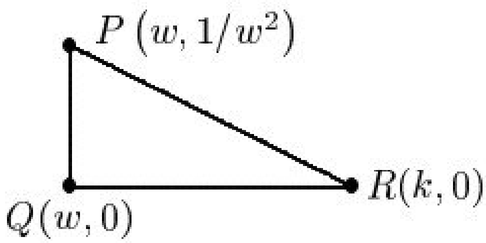
1 point is earned for the area in term or w and/or k

1 point is earned for: $\dfrac{dA}{dt}$ implicitly

4-5-5

1 point is earned for: $\left.\frac{dA}{dt}\right|_{w=5}$ using $\frac{dw}{dt} = 7$

1 point is earned for the conclusion



$A = \frac{1}{2}(k - w)\frac{1}{w^2} = \frac{1}{2}(\frac{3}{2}w - w)\frac{1}{w^2} = \frac{1}{4w}$
$\frac{dA}{dt} = -\frac{1}{4w^2}\frac{dw}{dt}$
$\left.\frac{dA}{dt}\right|_{w=5} = -\frac{1}{100} \cdot 7 = -0.07$

Therefore, area is decreasing.

0	1	2	3	4

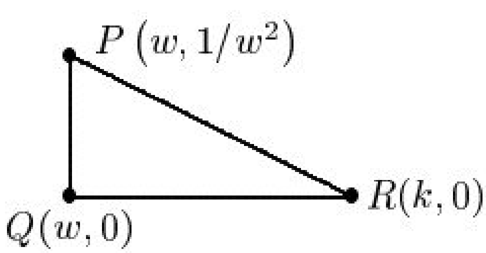
The student response earns all of the following points:

1 point is earned for the area in term or *w* and/or *k*

1 point is earned for: $\frac{dA}{dt}$ implicitly

1 point is earned for: $\left.\frac{dA}{dt}\right|_{w=5}$ using $\frac{dw}{dt} = 7$

1 point is earned for the conclusion



4-5-5

$$A = \tfrac{1}{2}(k - w)\frac{1}{w^2} = \tfrac{1}{2}\left(\tfrac{3}{2}w - w\right)\frac{1}{w^2} = \frac{1}{4w}$$
$$\frac{dA}{dt} = -\tfrac{1}{4w^2}\frac{dw}{dt}$$
$$\left.\frac{dA}{dt}\right|_{w=5} = -\tfrac{1}{100} \cdot 7 = -0.07$$

Therefore, area is decreasing.

4-5-5

55. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

For time $0 \leq t \leq 10$, water is flowing into a small tub at a rate given by the function F defined by $F(t) = \arctan\left(\frac{\pi}{2} - \frac{t}{10}\right)$. For time $5 \leq t \leq 10$, water is leaking from the tub at a rate given by the function L defined by $L(t) = 0.03\left(20t - t^2 - 75\right)$. Both $F(t)$ and $L(t)$ are measured in cubic feet per minute, and t is measured in minutes. The volume of water in the tub, in cubic feet, at time t minutes is given by $W(t)$.

(a) At time $t = 3$, there are 2.5 cubic feet of water in the tub. Write an equation for the locally linear approximation of W at $t = 3$, and use it to approximate the volume of water in the tub at time $t = 3.5$.

(b) Find $W''(8)$. Using correct units, interpret the meaning of $W''(8)$ in the context of the problem.

(c) Is there a time t, for $5 < t < 10$, at which the rate of change of the volume of water in the tub changes from positive to negative? Give a reason for your answer.

(d) The tub is in the shape of a rectangular box that is 0.5 foot wide, 4 feet long, and 3 feet deep. What is the rate of change of the depth of the water in the tub at time $t = 6$?

Part A

The second point may be earned if response contains an incorrect value for slope based on one computational error.

The third point does not require a simplified answer. Substitution of function values is required. Units are not required.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2	3

The student response accurately includes all three of the criteria below.

4-5-5

- ☐ slope
- ☐ equation for tangent line
- ☐ approximation

Solution:

$W(3) = 2.5$

$W'(3) = F(3) = 0.904089$

$y = W(3) + W'(3)(t-3) = 2.5 + 0.904(t-3)$

$W(3.5) \approx W(3) + W'(3)(3.5-3) = 2.952$ cubic feet

Part B

The second point requires a time reference, rate, and units. The point may be earned with an incorrect approximation based on one computational error and provided the behavior described is consistent with the sign of the approximation, the numerical value, and the time reference.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The student response accurately includes both of the criteria below.

- ☐ $W''(8)$
- ☐ interpretation with units

Solution:

$W''(8) = F'(8) - L'(8) = -0.183 \text{ (or } -0.182)$

At time $t = 8$ minutes, the rate at which the rate of change of the volume of water is changing by $-0.183 \text{ (or } -0.182)$ cubic feet per minute per minute.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

4-5-5

0	1

The student response accurately includes a correct answer with reason.

Solution:

$W'(t) = F(t) - L(t) = 0$ for $5 < t < 10$.

$W'(t) = 0 \quad \Rightarrow \quad t = 8.149287$

Yes, there is such a time t, because $W'(t) > 0$ for $5 < t < 8.149$ and $W'(t) < 0$ for $8.149 < t < 10$.

Part D

The first point is earned with $\frac{dV}{dt} = 2\frac{dh}{dt}$.

The third point may be earned with a maximum of one error in the evaluation of $\frac{dV}{dt}$.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2	3

The student response accurately includes all three of the criteria below.

- ☐ relationship between $\frac{dV}{dt}$ and $\frac{dh}{dt}$
- ☐ $\left.\frac{dV}{dt}\right|_{t=6}$
- ☐ answer

The volume of water in the tub is $V = (0.5)(4)h = 2h$, where h is the depth of water, in feet.

$$\frac{dV}{dt} = 2\frac{dh}{dt}$$

At time $t = 6$, $\left.\frac{dV}{dt}\right|_{t=6} = W'(t) = F(6) - L(6) = 0.500581$.

Therefore, $\left.\frac{dh}{dt}\right|_{t=6} = \frac{1}{2}\frac{dV}{dt} = 0.250$

4-5-5

56. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

For time $0 \leq t \leq 8$, people arrive at a venue for an outdoor concert at a rate modeled by the function A defined by $A(t) = 0.3\sin(1.9t) + 0.3\cos(0.6t) + 1.3$. For time $0 \leq t \leq 1$, no one leaves the venue, and for time $1 \leq t \leq 8$, people leave the venue at a rate modeled by the function L defined by $L(t) = 0.2\cos(1.9t) + 0.2\sin t + 0.8$. Both $A(t)$ and $L(t)$ are measured in hundreds of people per hour, and t is measured in hours. The number of people at the venue, in hundreds, at time t hours is given by $P(t)$.

(a) At time $t = 2$, there are 6 hundred people at the venue. Write an equation for the locally linear approximation of P at $t = 2$, and use it to approximate the number of people at the venue at time $t = 2.5$.

(b) Find $P''(5)$. Using correct units, interpret the meaning of $P''(5)$ in the context of the problem.

(c) Is there a time t, for $1 < t < 8$, at which the rate of change of the number of people at the venue changes from negative to positive? Give a reason for your answer.

(d) A rectangular "standing-only" section at the venue changes size as t increases in order to manage the flow of people. Let x represent the length, in feet, of the section, and let y represent the width, in feet, of the section. The length of the section is increasing at a rate of 6 feet per hour, and the width of the section is decreasing at a rate of 3 feet per hour. What is the rate of change of the area of the section with respect to time when $x = 16$ and $y = 10$? Indicate units of measure.

Part A

The second point may be earned if response contains an incorrect value for slope based on one computational error.

The third point **does not** require a simplified answer. Substitution of function values is required. An answer of 6.201 (or 6.200) alone does not earn the point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

4-5-5

0	1	2	3

The student response accurately includes all three of the criteria below.

- ☐ slope
- ☐ equation for tangent line
- ☐ approximation

Solution:

$P\left(2\right)=6$

$P'\left(2\right)=A\left(2\right)-L\left(2\right)=0.401484$

$y=P\left(2\right)+P'\left(2\right)\left(t-2\right)=6+0.401484\left(t-2\right)$

$P\left(2.5\right)\approx P\left(2\right)+P'\left(2\right)\left(2.5-2\right)=6.201$ (or 6.200) hundred people (≈ 620 OR ≈ 621 people)

Part B

The second point requires a time reference, rate, and units. The point may be earned with an incorrect approximation based on one computational error and provided the behavior described is consistent with the sign of the approximation, the numerical value, and the time reference.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2

The student response accurately includes both of the criteria below.

- ☐ $P''\left(5\right)$
- ☐ interpretation with units

Solution:

$P''\left(5\right)=A'\left(5\right)-L'\left(5\right)=-0.679$

At time $t=5$ hours, the rate of change of the number of people at the stage is changing at a rate of -0.679 hundred people (≈ -68 people) per hour per hour.

4-5-5

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1

The student response accurately includes a correct answer with reason.

Solution:

$P'(t) = A(t) - L(t) = 0$ for $1 < t < 8$.

$P'(t) = 0 \Rightarrow t = 5.811$ (or 5.810), $t = 6.548$ (or 6.547)

Yes, there is such a time t,because $P'(t) < 0$ for $5.811 < t < 6.548$ and $P'(t) > 0$ for $6.548 < t < 8$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2	3

The student response accurately includes all three of the criteria below.

- [ ] $\frac{dA}{dt} = x\frac{dy}{dt} + y\frac{dx}{dt}$
- [ ] $\frac{dx}{dt} = 6$ and $\frac{dy}{dt} = -3$
- [ ] answer with units

Solution:

Let x represent the length of the section, in feet, and let y represent the width of the section, in feet.

The area of the rectangular section is $A = xy$.

$$\frac{dA}{dt} = x\frac{dy}{dt} + y\frac{dx}{dt}$$

When $x = 16$ and $y = 10$, $\frac{dA}{dt} = x\frac{dy}{dt} + y\frac{dx}{dt} = (16)(-3) + (10)(6) = 12$.

Therefore, the rate of change of the area of the section with respect to time is 12 square feet per hour.

4-5-5

57. Oil is leaking from a pipeline on the surface of a lake and forms an oil slick whose volume increases at a constant rate of 2000 cubic centimeters per minute. The oil slick takes the form of a right circular cylinder with both its radius and height changing with time.(Note: The volume V of a right circular cylinder with radius r and height h is given by $V = \pi r^2 h$.)

(a) At the instant when the radius of the oil slick is 100 centimeters and the height is 0.5 centimeter, the radius is increasing at the rate of 2.5 centimeters per minute. At this instant, what is the rate of change of the height of the oil slick with respect to time, in centimeters per minute?

(b) A recovery device arrives on the scene and begins removing oil. The rate at which oil is removed is $R(t) = 400\sqrt{t}$ cubic centimeters per minute, where t is the time in minutes since the device began working. Oil continues to leak at the rate of 2000 cubic centimeters per minute. Find the time t when the oil slick reaches its maximum volume. Justify your answer.

(c) By the time the recovery device began removing oil, 60,000 cubic centimeters of oil had already leaked. Write, but do not evaluate, an expression involving an integral that gives the volume of oil at the time found in part (b).

Part A

1 point earned for $\frac{dV}{dt} = 2000$ and $\frac{dr}{dt} = 2.5$

When $r = 100$ cm and $h = 0.5$ cm, $\frac{dV}{dt} = 2000 \text{ cm}^3/\text{min}$ and $\frac{dr}{dt} = 2.5 \text{ cm}/\text{min}$.

2 point earned for the correct expression of $\frac{dV}{dt}$

$$\frac{dV}{dt} = 2\pi r \frac{dr}{dt} h + \pi r^2 \frac{dh}{dt}$$

1 point earned for the correct answer

$$2000 = 2\pi(100)(2.5)(0.5) + \pi(100)^2 \frac{dh}{dt}$$

$$\frac{dh}{dt} = 0.038 \text{ or } 0.039 \text{ cm}/\text{min}$$

0	1	2	3	4

The student response earns all of the following points:

1 point earned for $\frac{dV}{dt} = 2000$ and $\frac{dr}{dt} = 2.5$

4-5-5

When $r = 100$ cm and $h = 0.5$ cm, $\dfrac{dV}{dt} = 2000 \text{ cm}^3/\text{min}$ and $\dfrac{dr}{dt} = 2.5 \text{ cm}/\text{min}$.

2 point earned for the correct expression of $\dfrac{dV}{dt}$

$$\frac{dV}{dt} = 2\pi r \frac{dr}{dt} h + \pi r^2 \frac{dh}{dt}$$

1 point earned for the correct answer

$$2000 = 2\pi(100)(2.5)(0.5) + \pi(100)^2 \frac{dh}{dt}$$

$$\frac{dh}{dt} = 0.038 \text{ or } \ 0.039 \text{ cm}/\text{min}$$

Part B

1 point earned for $R(t) = 2000$

$$\frac{dV}{dt} = 2000 - R(t), \text{ so } \frac{dV}{dt} = 0 \text{ when } R(t) = 2000.$$

1 point earned for correct answer

This occurs when $t = 25$ minutes.

1 point earned for the justification

Since $\dfrac{dV}{dt} > 0$ for $0 < t < 25$ and $\dfrac{dV}{dt} < 0$ for $t > 25$, the oil slick reaches its maximum volume 25 minutes after the device begins working.

0	1	2	3 ✓

The student response earns all of the following points:

1 point earned for $R(t) = 2000$

$$\frac{dV}{dt} = 2000 - R(t), \text{ so } \frac{dV}{dt} = 0 \text{ when } R(t) = 2000.$$

1 point earned for correct answer

This occurs when $t = 25$ minutes.

1 point earned for the justification

4-5-5

Since $\frac{dV}{dt} > 0$ for $0 < t < 25$ and $\frac{dV}{dt} < 0$ for $t > 25$, the oil slick reaches its maximum volume 25 minutes after the device begins working.

Part C

1 point earned for the correct limits and initial condition

1 point earned for the integrand

The volume of oil, in cm^3, in the slick at time $t = 25$ minutes is given by $60,000 + \int_0^{25} (2000 - R(t))dt.$

0	1	2

The student response earns all of the following points:

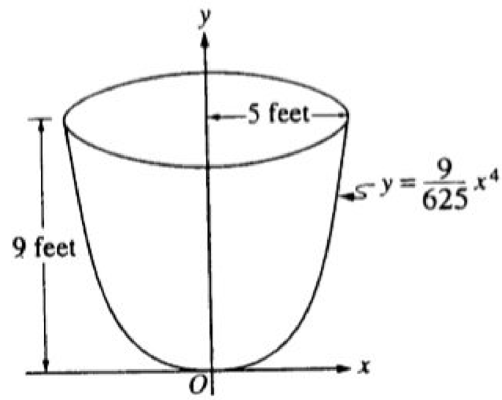
1 point earned for the correct limits and initial condition

1 point earned for the integrand

The volume of oil, in cm^3, in the slick at time $t = 25$ minutes is given by $60,000 + \int_0^{25} (2000 - R(t))dt.$

4-5-5

58.



An oil storage tank has the shape as shown above, obtained by revolving the curve $y = \frac{9}{625}x^4$ from $x = 0$ to $x = 5$ about the y-axis, where x and y are measured in feet.

Oil flows into the tank at the constant rate of 8 cubic feet per minute.
(a) Find the volume of the tank. Indicate units of measure.

(b) To the nearest minute, how long would it take to fill the tank if the tank was empty initially?

(c) Let h be the depth, in feet, of oil in the tank. How fast is the depth of oil in the tank increasing when $h = 4$? Indicate units of measure.

Part A

2 points are earned for the correct integral where one point is earned for the correct limits and

$\frac{\pi}{2\pi}$

$$\pi \int_0^9 (\ )dy, 2\pi \int_0^5 (\ )dx$$

and one point is earned for the correct integrand

Volume = $V = \pi = \int_0^9 \frac{25}{3}\sqrt{y}dy = 150\pi \text{ ft}^3$

(or 471.238 ft^3 or 471.239 ft^3)

1 point is earned for the correct answer with units

Note: not eligible for answer point unless first two points are earned

$$V = 2\pi = \int_0^5 x\left(9 - \frac{9}{625}x^4\right)dx = 150\pi \text{ ft}^3$$

[465, 475] [148π, 152π]

4-5-5

0	1	2	3

The student response earns all of the following points:

2 points are earned for the correct integral where one point is earned for the correct limits and

$$\frac{\pi}{2\pi}$$

$$\pi\int_0^9 (\;)dy, 2\pi\int_0^5 (\;)dx$$

and one point is earned for the correct integrand

Volume = $V = \pi = \displaystyle\int_0^9 \frac{25}{3}\sqrt{y}dy = 150\pi \text{ ft}^3$

(or 471.238 ft^3 or 471.239 ft^3)

1 point is earned for the correct answer with units

Note: not eligible for answer point unless first two points are earned

$$V = 2\pi = \int_0^5 x\left(9 - \frac{9}{625}x^4\right)dx = 150\pi \text{ ft}^3$$

[465, 475] [148π, 152π]

Part B

1 point is earned for the correct integer answer rounded up or down

$$\text{time} = \frac{\text{their Volume}}{\text{rate}} = \frac{150\pi}{8}$$

Therefore, 59 minutes

it is not necessary to indicate unit of measure

0	1

The student response earns all of the following points:

1 point is earned for the correct integer answer rounded up or down

4-5-5

$$\text{time} = \frac{\text{their Volume}}{\text{rate}} = \frac{150\pi}{8}$$

Therefore, 59 minutes

it is not necessary to indicate unit of measure

Part C

1 point is earned for the correct volume as definite integral using *h*

Finding $\frac{dV}{dt}$

$$V = \pi \int_0^h \frac{25}{3}\sqrt{y}dy = \frac{50\pi}{9}h^{3/2}$$

1 point is earned for the correct $\frac{dV}{dh}$

1 point is earned for the correct $\frac{dV}{dt} = \frac{dV}{dh} \cdot \frac{dh}{dt}$ by chain rule

1 point is earned for the correct answer $\frac{dV}{dt} = 8$

$$\frac{dV}{dt} = \frac{25}{3}\pi\sqrt{h}\frac{dh}{dt}$$
$$\frac{dV}{dt} = 8$$

when $h = 4, 8 = \frac{25}{3}\pi(2)\frac{dh}{dt}$

1 point is earned for the correct answer with units

Note: if *V* is linear, max 2/5 (0 - 0 - 1 - 1 - 0)

if *V* is constant, max 1/5 (0 - 0 - 0 - 1 - 0)

$$\frac{dh}{dt} = \frac{12}{25\pi} \text{ ft/min}$$

(or 0.152 ft/min or 0.153 ft/min)

0	1	2	3	4	5

The student response earns all of the following points:

1 point is earned for the correct volume as definite integral using *h*

4-5-5

Finding $\frac{dV}{dt}$

$$V = \pi \int_0^h \frac{25}{3}\sqrt{y}dy = \frac{50\pi}{9}h^{3/2}$$

1 point is earned for the correct $\frac{dV}{dh}$

1 point is earned for the correct $\frac{dV}{dt} = \frac{dV}{dh} \cdot \frac{dh}{dt}$ by chain rule

1 point is earned for the correct answer $\frac{dV}{dt} = 8$

$$\frac{dV}{dt} = \frac{25}{3}\pi\sqrt{h}\frac{dh}{dt}$$
$$\frac{dV}{dt} = 8$$

when $h = 4, 8 = \frac{25}{3}\pi(2)\frac{dh}{dt}$

1 point is earned for the correct answer with units

Note: if V is linear, max 2/5 (0 - 0 - 1 - 1 - 0)

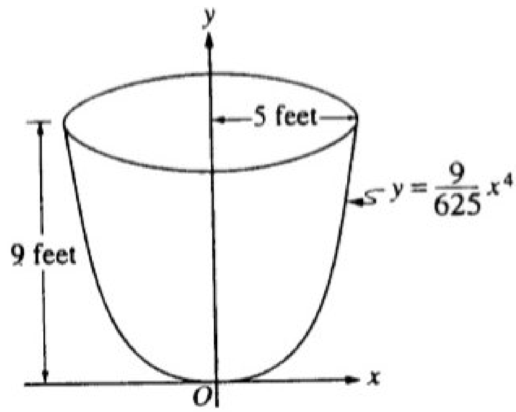
if V is constant, max 1/5 (0 - 0 - 0 - 1 - 0)

$\frac{dh}{dt} = \frac{12}{25\pi}$ ft/min

(or 0.152 ft/min or 0.153 ft/min)

4-5-5

59.



An oil storage tank has the shape as shown above, obtained by revolving the curve $y = \frac{9}{625}x^4$ from $x = 0$ to $x = 5$ about the y -axis, where x and y are measured in feet.

Oil weighing 50 pounds per cubic foot flowed into an initially empty tank at a constant rate of 8 cubic feet per minute. When the depth of oil reached 6 feet, the flow stopped.

(a) Let h be the depth, in feet, of oil in the tank. How fast was the depth of oil in the tank increasing when $h = 4$? Indicate units of measure.

(b) Find, to the nearest foot-pound, the amount of work required to empty the tank by pumping all of the oil back to the top of the tank.

Part A

1 point is earned for the correct volume as integral with h

$$V = \pi \int_0^h \frac{25}{3}\sqrt{y}\,dy$$

Finding $\frac{dV}{dt}$:

1 point is earned for the correct $\frac{dV}{dh}$

1 point is earned for the correct $\frac{dV}{dt} = \frac{dV}{dh} \cdot \frac{dh}{dt}$

$$\frac{dV}{dt} = \frac{25\pi}{3}\sqrt{h}\frac{dh}{dt}$$

1 point is earned for the correct answer $\frac{dV}{dt} = 8$

at $h = 4$, $8 = \frac{25\pi}{3}\sqrt{4}\frac{dh}{dt}$

1 point is earned for the correct answer with units

$$\frac{dh}{dt} = \frac{12}{25\pi}\ \mathrm{ft/min}$$

4-5-5

Note: If V is linear, max 2/5 (0 0 1 1 0)

If V is constant, max 1/5 (0 0 0 1 0)

0	1	2	3	4	5 ✓

The student response earns all of the following points:

1 point is earned for the correct volume as integral with h

$$V = \pi \int_0^h \frac{25}{3}\sqrt{y}\, dy$$

Finding $\frac{dV}{dt}$:

1 point is earned for the correct $\frac{dV}{dh}$

1 point is earned for the correct $\frac{dV}{dt} = \frac{dV}{dh} \cdot \frac{dh}{dt}$

$$\frac{dV}{dt} = \frac{25\pi}{3}\sqrt{h}\frac{dh}{dt}$$

1 point is earned for the correct answer $\frac{dV}{dt} = 8$

at $h = 4$, $8 = \frac{25\pi}{3}\sqrt{4}\frac{dh}{dt}$

1 point is earned for the correct answer with units

$$\frac{dh}{dt} = \frac{12}{25\pi}\ \text{ft/min}$$

Note: If V is linear, max 2/5 (0 0 1 1 0)

If V is constant, max 1/5 (0 0 0 1 0)

Part B

1 point is earned for the correct limits

$$W = 50\int_0^6 (9 - y)\left(\frac{25\pi}{3}\sqrt{y}\right) dy$$

2 points are earned for integrand where 1 point is earned for the force and 1 point is earned for the distance

4-5-5

$$W = 50\left(\frac{25\pi}{3}\right)\int_0^6 \left(9y^{\frac{1}{2}} - y^{\frac{3}{2}}\right)dy$$

$$W = 50\left(\frac{25\pi}{3}\right)\left(\frac{2}{3}\cdot 9y^{\frac{3}{2}} - \frac{2}{5}y^{\frac{5}{2}}\right)\Big|_0^6$$

1 point is earned for the correct answer

$W = 69{,}257.691$ ft-lbs to the nearest foot-pound 69,258 ft-lbs

(0/1 if no integrand points)

0	1	2	3	4

The student response earns all of the following points:

1 point is earned for the correct limits

$$W = 50\int_0^6 (9-y)\left(\frac{25\pi}{3}\sqrt{y}\right)dy$$

2 points are earned for integrand where 1 point is earned for the force and 1 point is earned for the distance

$$W = 50\left(\frac{25\pi}{3}\right)\int_0^6 \left(9y^{\frac{1}{2}} - y^{\frac{3}{2}}\right)dy$$

$$W = 50\left(\frac{25\pi}{3}\right)\left(\frac{2}{3}\cdot 9y^{\frac{3}{2}} - \frac{2}{5}y^{\frac{5}{2}}\right)\Big|_0^6$$

1 point is earned for the correct answer

$W = 69{,}257.691$ ft-lbs to the nearest foot-pound 69,258 ft-lbs

(0/1 if no integrand points)

4-5-5

60. Paint spills onto a floor in a circular pattern. The radius of the spill increases at a constant rate of 2.5 inches per minute. How fast is the area of the spill increasing when the radius of the spill is 18 inches?

(A) $5\pi \text{ in}^2/\text{min}$

(B) $36\pi \text{ in}^2/\text{min}$

(C) $45\pi \text{ in}^2/\text{min}$

(D) $90\pi \text{ in}^2/\text{min}$

Answer D

Correct. The area of the spill at time t is $A = \pi r^2$, where r is the radius at time t and is measured in inches and t is measured in minutes. The goal of this related rates problem is to find the value of $\frac{dA}{dt}$ at the moment when $r = 18$ inches and $\frac{dr}{dt} = 2.5$ inches per minute. By the chain rule,
$\frac{dA}{dt} = 2\pi r \frac{dr}{dt} = 2\pi(18)(2.5) = 90\pi \text{ in}^2/\text{min}.$

61. A particle moves along the curve $y = \frac{15}{x^2+1.3^x}$ for $x > 0$. The x-coordinate of the particle changes at a constant rate of 3 units per second. At the instant when the y-coordinate of the particle is 2, what is the rate of change of the y-coordinate of the particle, in units per second?

(A) -0.466

(B) -0.787

(C) -1.397

(D) -4.190

Answer D

Correct. By the chain rule, $\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}$. The y-coordinate of the particle is 2 at the instant when
$2 = \frac{15}{x^2+1.3^x} \Rightarrow x = 2.373978.$
At this instant $\frac{dx}{dt} = 3$ and therefore $\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt} = \left(\frac{d}{dx}\left(\frac{15}{x^2+1.3^x}\right)\Big|_{x=2.373978} \right) \cdot 3 = -4.190.$

4-5-5

62. A spherical snowball is melting in such a way that it maintains its shape. The volume of the snowball is decreasing at a constant rate of 5 cubic inches per minute. At the instant when the radius of the snowball is decreasing at a rate of $\frac{3}{5}$ inch per minute, what is the radius of the snowball, in inches? (The volume V of a sphere with radius r is $V = \frac{4}{3}\pi r^3$.)

(A) $\sqrt{\frac{5}{4\pi}}$

(B) $\frac{25}{12\pi}$

(C) $\frac{5}{\sqrt{12\pi}}$ ✓

(D) $\frac{125}{36\pi}$

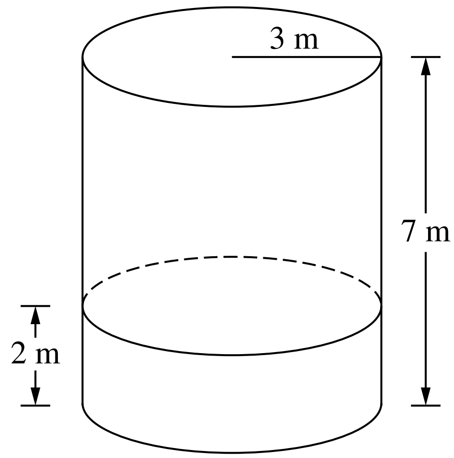
Answer C

Correct. The volume of the snowball is $V = \frac{4}{3}\pi r^3$, where r is the radius at time t and is measured in inches and t is measured in minutes. The goal of this related rates problem is to find the value of r at the instant when $\frac{dV}{dt} = -5$ cubic inches per minute and $\frac{dr}{dt} = -\frac{3}{5}$ inch per minutes. Both rates of change are negative because the volume and radius are decreasing. By the chain rule,

$$\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt} \Rightarrow -5 = 4\pi r^2 \left(-\frac{3}{5}\right) \Rightarrow r^2 = \frac{25}{12\pi} \Rightarrow r = \sqrt{\frac{25}{12\pi}} = \frac{5}{\sqrt{12\pi}}.$$

4-5-5

63.



A water tank standing on level ground is in the shape of a right circular cylinder, as shown. Water is flowing into the tank at the rate of 2 cubic meters per minute. The tank has a radius of 3 meters and a height of 7 meters. At what rate, in meters per minute, is the depth of the water changing when the water is 2 meters deep? (The volume V of a right circular cylinder with radius r and height h is $V = \pi r^2 h$.)

(A) 0.071 ✓

(B) 0.095

(C) 0.106

(D) 0.222

Answer A

Correct. The volume of the water is $V = \pi r^2 h = \pi(3)^2 h = 9\pi h$ with $\frac{dV}{dt} = 2$. The question asks for the value of $\frac{dh}{dt}$ at the moment that $h = 2$.

$\frac{dV}{dt} = 9\pi \frac{dh}{dt} \Rightarrow \frac{dh}{dt} = \frac{1}{9\pi} \frac{dV}{dt} = \frac{2}{9\pi} = 0.071$

4-5-5

64. A thin rectangular sheet of metal expands in a hot oven. At the instant when the width of the rectangular sheet of metal is 200 millimeters and the length is 300 millimeters, the width is increasing at a rate of 2 millimeters per minute and the length is increasing at a rate of 3 millimeters per minute. What is the rate of change of the area, in square millimeters per minute, of the rectangular sheet of metal at that instant?

(A) 5

(B) 6

(C) 600

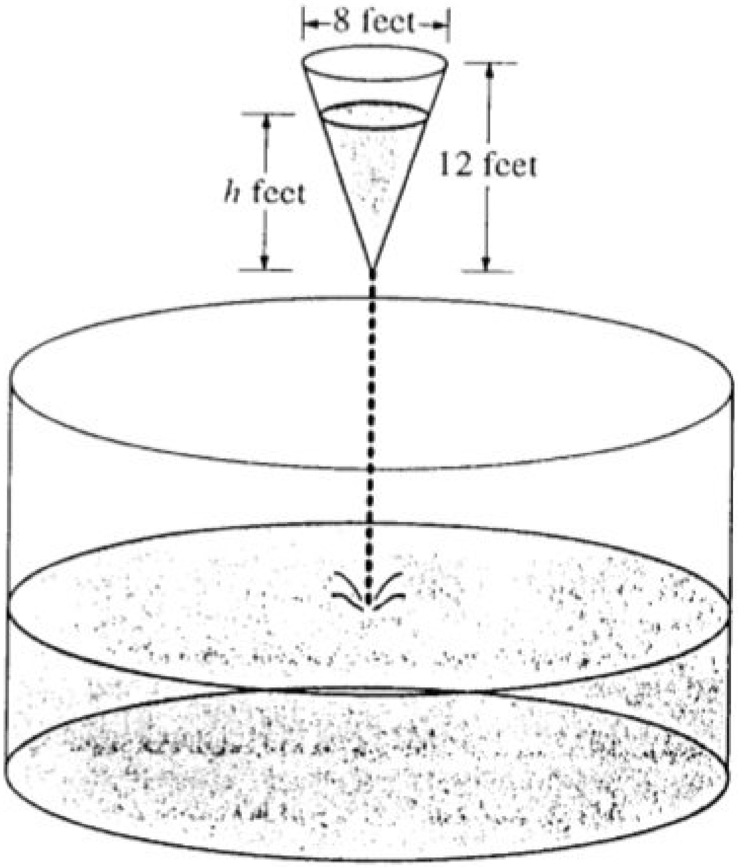
(D) 1200

Answer D

Correct. The area of the sheet is given by $A = wh$, where both the width w and the length h are functions of time. At the instant described, $w = 200$ and $\frac{dw}{dt} = 2$, while $h = 300$ and $\frac{dh}{dt} = 3$. The product rule and chain rule can be used to determine that at this instant $\frac{dA}{dt} = \frac{dw}{dt} \cdot h + w \cdot \frac{dh}{dt} = (2)(300) + (200)(3) = 600 + 600 = 1200$.

4-5-5

65.



As shown in the figure above, water is draining from a conical tank with height 12 feet and diameter 8 feet into a cylindrical tank that has a base with area 400π square feet. The depth h, in feet, of the water in the conical tank is changing at the rate of $(h - 12)$ feet per minute. (The volume V of a cone with radius r and height h is $V = \frac{1}{3}\pi r^2 h$.)

(a) Write an expression for the volume of water in the conical tank as a function of h.

(b) At what rate is the volume of water in the conical tank changing when $h = 3$? Indicate units of measure.

(c) Let y be the depth, in feet, of the water in the cylindrical tank. At what rate is y changing when $h = 3$? Indicate units of measure.

Part A

1 point is earned for the correct answer $r = \frac{1}{3}h$

$$\frac{r}{h} = \frac{4}{12} = \frac{1}{3}$$
$$r = \frac{1}{3}h$$

1 point is earned for correctly showing V as a function of h

$$V = \frac{1}{3}\pi\left(\frac{1}{3}h\right)^2 h = \frac{\pi h^3}{27}$$

4-5-5

Note: 0/2 if r constant

0	1	2

The student response earns all of the following points:

1 point is earned for the correct answer $r = \frac{1}{3}h$

$$\frac{r}{h} = \frac{4}{12} = \frac{1}{3}$$
$$r = \frac{1}{3}h$$

1 point is earned for correctly showing V as a function of h

$$V = \frac{1}{3}\pi\left(\frac{1}{3}h\right)^2 h = \frac{\pi h^3}{27}$$

Note: 0/2 if r constant

Part B

1 point is earned for correctly solving $\frac{dV}{dt}$ using the chain rule

$$\frac{dV}{dt} = \frac{\pi h^2}{9}\frac{dh}{dt}$$

1 point is earned for the correct answer with

$$= \frac{\pi h^2}{9}(h - 12) = -9\pi$$

1 point is earned for correctly solving for $\frac{dV}{dt}$ and gives answer with unit

V is decreasing at $9\pi \text{ ft}^3/\text{min}$

Note: 0/1 if $\frac{dV}{dt} > 0$

0	1	2	3

The student response earns all of the following points:

1 point is earned for correctly solving $\frac{dV}{dt}$ using the chain rule

4-5-5

$$\frac{dV}{dt} = \frac{\pi h^2}{9}\frac{dh}{dt}$$

1 point is earned for the correct answer with

$$= \frac{\pi h^2}{9}(h - 12) = -9\pi$$

1 point is earned for correctly solving for $\frac{dV}{dt}$ and gives answer with unit

V is decreasing at $9\pi\ \mathrm{ft}^3/\ \mathrm{min}$

Note: 0/1 if $\frac{dV}{dt} > 0$

Part C

1 point is earned for correctly defining W as a function of y

Let W = volume of water in cylindrical tank

$$W = 400\pi y;\ \frac{dW}{dt} = 400\pi\frac{dy}{dt}$$
$$400\pi\frac{dy}{dt} = 9\pi$$

1 point is earned for the correct answer for $\frac{dW}{dt} = 400\pi\frac{dy}{dt}$

1 point is earned for correctly identifying $\frac{dW}{dt} = \left|\frac{dV}{dt}\right|$ or $-\frac{dV}{dt}$

1 point is earned for correctly solving for $\frac{dy}{dt}$ and gives answer with unit

y is increasing at $\frac{9}{400}$ ft/min

0	1	2	3	4 ✓

The student response earns all of the following points:

1 point is earned for correctly defining W as a function of y

Let W = volume of water in cylindrical tank

4-5-5

$W = 400\pi y; \dfrac{dW}{dt} = 400\pi \dfrac{dy}{dt}$

$400\pi \dfrac{dy}{dt} = 9\pi$

1 point is earned for the correct answer for $\dfrac{dW}{dt} = 400\pi \dfrac{dy}{dt}$

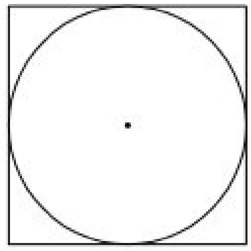
1 point is earned for correctly identifying $\dfrac{dW}{dt} = \left|\dfrac{dV}{dt}\right|$ or $-\dfrac{dV}{dt}$

1 point is earned for correctly solving for $\dfrac{dy}{dt}$ and gives answer with unit

y is increasing at $\frac{9}{400}$ ft/min

4-5-5

66.



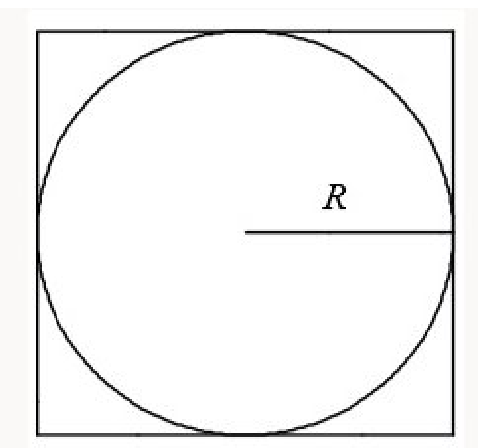
A *circle is inscribed in a square as shown in the figure above. The circumference of the circle is increasing at a constant rate of 6 inches per second. As the circle expands, the square expands to maintain the condition of tangency.* (Note: A circle with radius r has circumference $C = 2\pi r$ and area $A = \pi r^2$)

(a) Find the rate at which the perimeter of the square is increasing. Indicate units of measure.

(b) At the instant when the area of the circle is 25π square inches, find the rate of increase in the area enclosed between the circle and the square. Indicate units of measure.

Part A

1 point is earned for correctly expressing the perimeter of square in terms of circle data



1 point is earned for correctly differentiating to find $\frac{dP}{dt}$

$P = 8R$

$\frac{dP}{dt} = 8\frac{dR}{dt}$

1 point is earned for the correct answer $\frac{dC}{dt} = 6$

$6 = \frac{dC}{dt} = 2\pi\frac{dR}{dt}$

1 point is earned for correctly solving for $\frac{dP}{dt}$

$\frac{dR}{dt} = \frac{3}{\pi}; \frac{dP}{dt} = \frac{24}{\pi}$ inches/second

≈ 7.639 inches/second

4-5-5

1 point is earned for correct units for answers in BOTH (a) and (b)

Units: inches/second in (a)

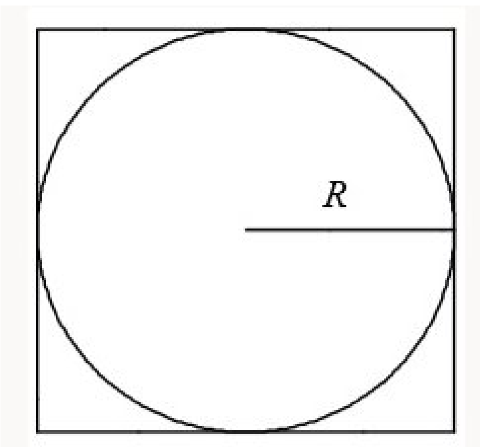
square inches/second in (b)

0/1 if no use of $\frac{dC}{dt}$

0	1	2	3	4

The student response earns all of the following points:

1 point is earned for correctly expressing the perimeter of square in terms of circle data



1 point is earned for correctly differentiating to find $\frac{dP}{dt}$

$P = 8R$

$\frac{dP}{dt} = 8\frac{dR}{dt}$

1 point is earned for the correct answer $\frac{dC}{dt} = 6$

$6 = \frac{dC}{dt} = 2\pi\frac{dR}{dt}$

1 point is earned for correctly solving for $\frac{dP}{dt}$

$\frac{dR}{dt} = \frac{3}{\pi}; \frac{dP}{dt} = \frac{24}{\pi}$ inches/second

≈ 7.639 inches/second

1 point is earned for correct units for answers in BOTH (a) and (b)

Units: inches/second in (a)

4-5-5

square inches/second in (b)

0/1 if no use of $\frac{dC}{dt}$

Part B

1 point is earned for the correct area of region between the square and circle

$$\text{Area} = 4R^2 - \pi R^2$$

1 point is earned for the correct derivative of area with respect to t

1/1: for derivatives of areas of square and circle separately

0/1: if area of either square or circle is linear

$$\frac{d(\text{Area})}{dt} = 8R\frac{dR}{dt} - 2\pi R\frac{dR}{dt}$$
$$= (4-\pi)2R\frac{dR}{dt}$$

1 point is earned for correctly using $R = 5$ in derivative

$$\text{Area of circle} = 25\pi = \pi R^2$$
$$R = 5$$

1 point is earned for the correct answer

$$\frac{d(\text{Area})}{dt} = \frac{120}{\pi} - 30 \text{ inches}^2/\text{second}$$
$$= (4-\pi)\,\frac{30}{\pi} \text{ inches}^2/\text{second}$$
$$\approx 8.197 \text{ inches}^2/\text{second}$$

1 point is earned for correct units for answers in both (a) and (b)

Units: inches/second in (a)

square inches/second in (b)

0	1	2	3	4	5

The student response earns all of the following points:

1 point is earned for the correct area of region between the square and circle

4-5-5

$$\text{Area} = 4R^2 - \pi R^2$$

1 point is earned for the correct derivative of area with respect to t

1/1: for derivatives of areas of square and circle separately

0/1: if area of either square or circle is linear

$$\begin{aligned}\frac{d(\text{Area})}{dt} &= 8R\frac{dR}{dt} - 2\pi R\frac{dR}{dt} \\ &= (4-\pi)2R\frac{dR}{dt}\end{aligned}$$

1 point is earned for correctly using $R = 5$ in derivative

$$\begin{aligned}\text{Area of circle} = \ 25\pi &= \pi R^2 \\ R &= 5\end{aligned}$$

1 point is earned for the correct answer

$$\begin{aligned}\frac{d(\text{Area})}{dt} &= \frac{120}{\pi} - 30 \text{ inches}^2/\text{second} \\ &= (4-\pi)\,\frac{30}{\pi} \text{ inches}^2/\text{second} \\ &\approx 8.197 \text{ inches}^2/\text{second}\end{aligned}$$

1 point is earned for correct units for answers in both (a) and (b)

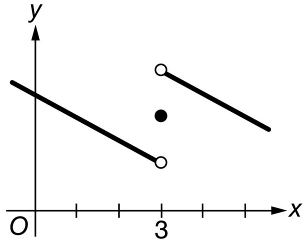
Units: inches/second in (a)

square inches/second in (b)

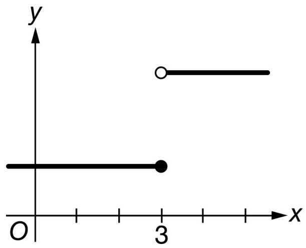
4-5-5

67. If f is a function that has a removable discontinuity at $x = 3$, which of the following could be the graph of f ?

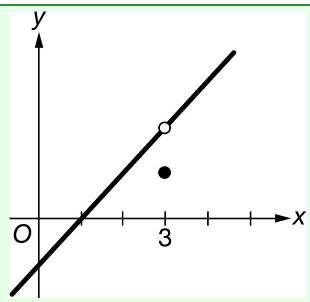
(A)



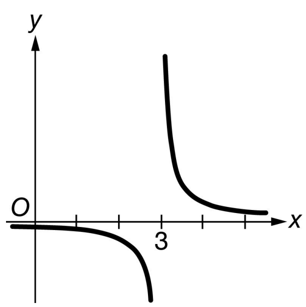
(B)



(C)



(D)



Answer C

Correct. A removable discontinuity occurs at $x = c$ if $\lim_{x \to c} f(x)$ exists, but $f(c)$ does not exist or is not equal to the value of the limit. This graph could be the graph of f since $\lim_{x \to 3} f(x)$ exists but is not equal to $f(3)$.

4-5-5

68. At a certain height, a tree trunk has a circular cross section. The radius $R(t)$ of that cross section grows at a rate modeled by the function

$$\frac{dR}{dt} = \frac{1}{16}\left(3 + \sin\left(t^2\right)\right) \text{ centimeters per year}$$

for $0 \le t \le 3$, where time t is measured in years. At time $t = 0$, the radius is 6 centimeters. The area of the cross section at time t is denoted by $A(t)$.

(a) Write an expression, involving an integral, for the radius $R(t)$ for $0 \le t \le 3$. Use your expression to find $R(3)$.

(b) Find the rate at which the cross-sectional area $A(t)$ is increasing at time $t = 3$ years. Indicate units of measure.

(c) Evaluate $\int_0^3 A'(t)dt$. Using appropriate units, interpret the meaning of that integral in terms of cross-sectional area.

Part A

1 point is earned for the integral

1 point is earned for the expression for $R(t)$

1 point is earned for $R(3)$

$$R(t) = 6 + \int_0^t \frac{1}{16}\left(3 + \sin\left(x^2\right)\right)dx$$

$R(3) = 6.610$ or 6.611

0	1	2	3

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the expression for $R(t)$

1 point is earned for $R(3)$

$$R(t) = 6 + \int_0^t \frac{1}{16}\left(3 + \sin\left(x^2\right)\right)dx$$

$R(3) = 6.610$ or 6.611

Part B

1 point is earned for expression of $A(t)$

4-5-5

1 point is earned for expression of $A'(t)$

1 point is earned for answering with units

$$A(t) = \pi(R(t))^2$$
$$A'(t) = 2\pi R(t)R'(t)$$
$$A'(3) = 8.858 \text{ cm}^2/\text{year}$$

0	1	2	3

The student response earns all of the following points:

1 point is earned for expression of $A(t)$

1 point is earned for expression of $A'(t)$

1 point is earned for answering with units

$$A(t) = \pi(R(t))^2$$
$$A'(t) = 2\pi R(t)R'(t)$$
$$A'(3) = 8.858 \text{ cm}^2/\text{year}$$

Part C

1 point is earned for using Fundamental Theorem of Calculus

1 point is earned for value of $\int_0^3 A'(t)dt$

1 point is earned for meaning of $\int_0^3 A'(t)dt$

$\int_0^3 A'(t)dt = A(3) - A(0) = 24.200$ or 24.201

From time $t = 0$ to $t = 3$ years, the cross-sectional area grows by 24.201 square centimeters.

4-5-5

0	1	2	3

The student response earns all of the following points:

1 point is earned for using Fundamental Theorem of Calculus

1 point is earned for value of $\int_0^3 A'(t)dt$

1 point is earned for meaning of $\int_0^3 A'(t)dt$

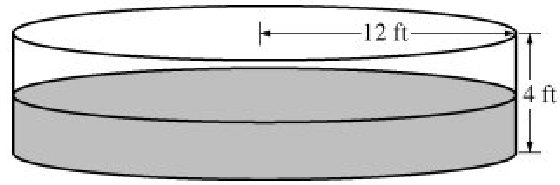
$\int_0^3 A'(t)dt = A(3) - A(0) = 24.200$ or 24.201

From time $t = 0$ to $t = 3$ years, the cross-sectional area grows by 24.201 square centimeters.

4-5-5

69.

t	0	2	4	6	8	10	12
$P(t)$	0	46	53	57	60	62	63



The figure above shows an aboveground swimming pool in the shape of a cylinder with a radius of 12 feet and a height of 4 feet. The pool contains 1000 cubic feet of water at time $t = 0$. During the time interval $0 \leq t \leq 12$ hours, water is pumped into the pool at the rate $P(t)$ cubic feet per hour. The table above gives values of $P(t)$ for selected values of t. During the same time interval, water is leaking from the pool at the rate $R(t)$ cubic feet per hour, where $R(t) = 25e^{-0.05t}$. (Note: The volume V of a cylinder with radius r and height h is given by $V = \pi r^2 h$.)

(a) Use a midpoint Riemann sum with three subintervals of equal length to approximate the total amount of water that was pumped into the pool during the time interval $0 \leq t \leq 12$ hours. Show the computations that lead to your answer.

(b) Calculate the total amount of water that leaked out of the pool during the time interval $0 \leq t \leq 12$ hours.

(c) Use the results from parts (a) and (b) to approximate the volume of water in the pool at time $t = 12$ hours. Round your answer to the nearest cubic foot.

(d) Find the rate at which the volume of water in the pool is increasing at time $t = 8$ hours. How fast is the water level in the pool rising at $t = 8$ hours? Indicate units of measure in both answers.

Part A

1 point is earned for midpoint sum

1 point is earned for the answer

$$\int_0^{12} P(t)dt \approx 46 \cdot 4 + 57 \cdot 4 + 62 \cdot 4 = 660 \text{ ft}^3$$

0	1	2

The student response earns all of the following points:

1 point is earned for midpoint sum

1 point is earned for the answer

$$\int_0^{12} P(t)dt \approx 46 \cdot 4 + 57 \cdot 4 + 62 \cdot 4 = 660 \text{ ft}^3$$

4-5-5

Part B

1 point is earned for the integral

1 point is earned for the answer

$$\int_0^{12} R(t)dt = 225.594 \text{ ft}^3$$

0	1	2

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the answer

$$\int_0^{12} R(t)dt = 225.594 \text{ ft}^3$$

Part C

1 point is earned for the answer

$$1000 + \int_0^{12} P(t)dt - \int_0^{12} R(t)dt = 1434.406$$

At time $t = 12$ hours, the volume of water in the pool is approximately 1434 ft^3.

0	1

The student response earns all of the following points:

1 point is earned for the answer

4-5-5

$$1000+\int_0^{12} P(t)dt-\int_0^{12} R(t)dt=1434.406$$

At time $t = 12$ hours, the volume of water in the pool is approximately 1434 ft^3.

Part D

1 point is earned for $V'(8)$

1 point is earned for equation relating $\frac{dV}{dt}$ and $\frac{dh}{dt}$

1 point is earned for $\left.\frac{dh}{dt}\right|_{t=8}$

1 point is earned for units of $\mathrm{ft}^3/\mathrm{hr}$ and ft/ hr

$$V'(t)=P(t)-R(t)$$
$$V'(8)=P(8)-R(8)=60-25e^{-0.4}=43.241 \text{ or } 43.242 \ \mathrm{ft}^3/\mathrm{hr}$$

$$V=\pi(12)^2h$$
$$\frac{dV}{dt}=144\pi\frac{dh}{dt}$$
$$\left.\frac{dh}{dt}\right|_{t=8}=\frac{1}{144\pi}\cdot\left.\frac{dV}{dt}\right|_{t=8}=0.095 \text{ or } 0.096 \ \mathrm{ft}/\mathrm{hr}$$

0	1	2	3	4

The student response earns all of the following points:

1 point is earned for $V'(8)$

1 point is earned for equation relating $\frac{dV}{dt}$ and $\frac{dh}{dt}$

4-5-5

1 point is earned for $\left.\frac{dh}{dt}\right|_{t=8}$

1 point is earned for units of ft^3/hr and ft/ hr

$V'(t) = P(t) - R(t)$
$V'(8) = P(8) - R(8) = 60 - 25e^{-0.4} = 43.241 \text{ or } 43.242 \text{ ft}^3/\text{hr}$

$V = \pi(12)^2 h$
$\frac{dV}{dt} = 144\pi \frac{dh}{dt}$
$\left.\frac{dh}{dt}\right|_{t=8} = \frac{1}{144\pi} \cdot \left.\frac{dV}{dt}\right|_{t=8} = 0.095 \text{ or } 0.096 \text{ ft/hr}$

4-5-5

70. The wind chill is the temperature, in degrees Fahrenheit (°F), a human feels based on the air temperature, in degrees Fahrenheit, and the wind velocity v, in miles per hour (mph). If the air temperature is 32°F, then the wind chill is given by $W(v) = 55.6 - 22.1v^{0.16}$ and is valid for $5 \le v \le 60$.

(a) Find $W'(20)$. Using correct units, explain the meaning of $W'(20)$ in terms of the wind chill.

(b) Find the average rate of change of W over the interval $5 \le v \le 60$. Find the value of v at which the instantaneous rate of change of W is equal to the average rate of change of W over the interval $5 \le v \le 60$.

(c) Over the time interval $0 \le t \le 4$ hours, the air temperature is a constant 32°F. At time $t = 0$, the wind velocity is $v = 20$ mph. If the wind velocity increases at a constant rate of 5 mph per hour, what is the rate of change of the wind chill with respect to time at $t = 3$ hours? Indicate units of measure.

Part A

1 point is earned for the value

1 point is earned for the explanation

$$W'(20) = -22.1 \cdot 0.16 \cdot 20^{-0.84} = -0.285 \text{ or } -0.286$$

When $v = 20$ mph, the wind chill is decreasing at 0.286 $^{\circ}$F /mph.

1 point is earned for units in (a) and (c)

Units of °F/mph in (a) and °F/hr in (c)

0	1	2	3

The student response earns all of the following points:

1 point is earned for the value

1 point is earned for the explanation

$$W'(20) = -22.1 \cdot 0.16 \cdot 20^{-0.84} = -0.285 \text{ or } -0.286$$

When $v = 20$ mph, the wind chill is decreasing at 0.286 $^{\circ}$F /mph.

1 point is earned for units in (a) and (c)

Units of °F/mph in (a) and °F/hr in (c)

Part B

4-5-5

1 point is earned for the average rate of change.

1 point is earned for $W'(v)$ = average rate of change

1 point is earned for the value of v.

The average rate of change of W over the interval $5 \leq v \leq 60$ is $\dfrac{W(60)-W(5)}{60-5}$ = -0.253 or -0.254.

$W'(v) = \dfrac{W(60)-W(5)}{60-5}$ when $v = 23.011$.

0	1	2	3 ✓

The student response earns all of the following points:

1 point is earned for the average rate of change.

1 point is earned for $W'(v)$ = average rate of change

1 point is earned for the value of v.

The average rate of change of W over the interval $5 \leq v \leq 60$ is $\dfrac{W(60)-W(5)}{60-5}$ = -0.253 or -0.254.

$W'(v) = \dfrac{W(60)-W(5)}{60-5}$ when $v = 23.011$.

Part C

1 point is earned for $\dfrac{dv}{dt} = 5$

1 point is earned if uses $v(3) = 35$ or uses $v(5) = 20 + 5t$

1 point is earned for answer

4-5-5

$$\left.\frac{dW}{dt}\right|_{t=3} = \left.\left(\frac{dW}{dv} \cdot \frac{dv}{dt}\right)\right|_{t=3} = W'(35) \cdot 5 = -0.892^{\circ}\mathrm{F/hr}$$

OR

$$W = 55.6 - 22.1(20 + 5t)^{0.16}$$

$\left.\frac{dW}{dt}\right|_{t=3}$ = -0.892 °F/hr

1 point is earned for units in (a) and (c)

Units of °F/mph in (a) and °F/hr in (c)

0	1	2	3	4

The student response earns all of the following points:

1 point is earned for $\frac{dv}{dt} = 5$

1 point is earned if uses $v(3) = 35$ or uses $v(5) = 20 + 5t$

1 point is earned for answer

$$\left.\frac{dW}{dt}\right|_{t=3} = \left.\left(\frac{dW}{dv} \cdot \frac{dv}{dt}\right)\right|_{t=3} = W'(35) \cdot 5 = -0.892^{\circ}\mathrm{F/hr}$$

OR

$$W = 55.6 - 22.1(20 + 5t)^{0.16}$$

$\left.\frac{dW}{dt}\right|_{t=3}$ = -0.892 °F/hr

1 point is earned for units in (a) and (c)

4-5-5

Units of °F/mph in (a) and °F/hr in (c)